

TECHNICAL MANUAL

AF22e

UV FLUORESCENT SULFUR DIOXIDE ANALYZER

- FEBRUARY 2020 -



WARNING

The information in this documentation is subject to change without notice.

ENVEA all rights reserved.

This document does not represent a commitment under part of ENVEA

Warranty

Defects that fall under warranty

The seller shall undertake to remedy any operational malfunction resulting from a manufactured or material defect within the limits of the provision below.

The seller shall not be liable in the case of a defect caused either by materials supplied by the buyer or by a design imposed by the buyer.

Any warranty is also invalid in the case of damage resulting from normal wear and tear, accident, disaster, misuse, fault or negligence of or by Buyer, causes external to the Products such as, but not limited to, unauthorized repairs or part replacement, electrical power surges, improper storage of the Product, use of the Product in a manner for which it was not designed.

Duration and starting point of the warranty

Unless otherwise stipulated, the warranty period shall have a duration of twelve months from the date of delivery within the meaning of article 6 paragraph 2 of the «ENVEA: INTERNATIONAL GENERAL TERMS AND CONDITIONS OF SALES», even if the shipment or assembly is postponed for any reason outside the seller's control.

Unless agreed upon by both parties, the repair, modification or replacement of parts during the warranty period will not extend or renew the original equipment warranty period.

Buyer's obligation

In order to file a claim under warranty, the buyer must notify the seller immediately in writing of any defect in the equipment and supply evidence in proof thereof. The buyer must provide the seller with the opportunity to observe and remedy the said defects. In addition, the buyer must not carry out any repairs or have repairs made by a third party without the written agreement of the seller.

The buyer is required to check the equipment as soon as possible upon receipt and acceptance and no later than eight days following receipt. Failing to do so might invalidate any claims made later regarding a declared defect.

Any installation, maintenance, repair, service of the product performed by any person or entity other than seller without seller's prior written approval, or any use of replacement parts not supplied by seller, shall immediately void and cancel all warranties with respect to the affected product.

Exercising the warranty

Once notified of a defect, the seller shall be responsible for remedying the defect at its own expense. The seller, however, reserves the right to modify the mechanisms of the equipment as needed to comply with its obligations.

The work to satisfy the warranty obligation shall be carried out, principally, in the seller's workshop after the buyer has returned the equipment or the defective parts to the seller for the purposes of repair or replacement, whichever the seller deems best.

However, if the nature of the equipment is such that the repair has to be carried out at the location where it was installed, the seller shall only be responsible for the on-site labor costs involved in direct service of the analyzer itself. The buyer is responsible for the cost of any additional measures needed to provide unrestricted access.

The cost of transport of the equipment or the defective parts, as well as the return of the repaired or replaced equipment or part, shall be borne by the buyer. In the case of on-site repair, the buyer shall be responsible for any travelling and accommodation expenses of the seller's representative.

Defective parts replaced free of charge, under warranty, must be returned to the seller and shall become its property once again.

SUMMARY

CHAPTER 1. GENERAL – CHARACTERISTICS

1.1.	GENERAL	1–3
1.2.	CHARACTERISTICS	1–9

CHAPTER 2. PRINCIPLE OF OPERATION

2.1	THEORETICAL BASIS	2–3
2.2	MEASUREMENT PRINCIPLE	2–6
2.3	GENERAL PRINCIPLE DIAGRAMS	2–7
2.4	SAMPLING AND ANALYSIS	2–9
2.5	SIMPLIFIED CALCULATION DESCRIPTION	2–11
2.6	PROGRAMMING THE RESPONSE TIME	2–12
2.7	ELECTRONIC ARCHITECTURE	2–13
2.8	NETWORK LINK AND USB PORTS	2–14

CHAPTER 3. OPERATING INSTRUCTIONS

3.1.	INITIAL STARTUP	3–3
3.2.	PROGRAMMING THE AF22E	3–9
3.3.	BROWSING ERGONOMY	3–13
3.4.	ANALYZER FUNCTION DESCRIPTION	3–18
3.5.	CALIBRATION	3–38

CHAPTER 4. PREVENTIVE MAINTENANCE

4.1	SAFETY INSTRUCTIONS	4-3
4.2	MAINTENANCE CALENDAR	4-4
4.3	MAINTENANCE OPERATION SHEETS	4-4
4.4	KITS AND SPARE PARTS SET FOR AF22e MAINTENANCE	4-17

CHAPTER 5. CORRECTIVE MAINTENANCE

List of Figures

Figure 1-1 – ENVEA AF22e (with screen)	1-2
Figure 1-2 – ENVEA AF22e* (without screen)	1-2
Figure 1-3 – Color screen fitted with touch-screen	1-3
Figure 1-4 – Rear panel	1-5
Figure 1-5 – AF22e internal view	1-7
Figure 1-6 – Links between units	1-10
Figure 1-7 – AF22e dimensions (in mm)	1-11
Figure 2-1 – Diagram showing molecule energy levels	2-5
Figure 2-2 – General principle diagram of standard AF22e	2-7
Figure 2-3 – General principle diagram of AF22e with permeation bench option	2-8
Figure 2-4 – Hydrocarbon molecule filtration	2-9
Figure 2-5 – Electronic architecture	2-14
Figure 3-1 – Fluid and electrical connections	3-3
Figure 3-2 – Installation of the sample gas inlet	3-4
Figure 3-3 – Example of a pressurized gas connection	3-39
Figure 3-4 – Typical calibration device diagram	3-44
Figure 4-1 – Sample inlet dust filter replacement	4-5
Figure 4-2 – Internal zero filter replacement	4-8
Figure 4-3 – KNF pump head dismantling	4-10
Figure 4-4 – Diaphragm replacement	4-11
Figure 4-5 – Valves and O-rings replacement	4-12
Figure 4-6 – Pump reassembly	4-13
Figure 4-7 – UV source assembly dismantling	4-14
Figure 4-8 – UV lamp change	4-15
Figure 4-9 – UV source reassembly	4-16
Figure 5-1 – Interconnection board diagram	5-11
Figure 5-2 – Controller board diagram (above view)	5-12
Figure 5-3 – Power supply board diagram	5-13
Figure 5-4 – Measurement board diagram	5-15
Figure 5-5 – Photodiode board diagram (above view)	5-16
Figure 5-6 – PM preamplifier board diagram	5-17
Figure 5-7 – UV lamp power board diagram	5-18
Figure 5-8 – ARM20 board	5-19

List of Tables

Table 5–1 – List of faults and corrective actions	5–4
Table 5–2 – Interconnection board configuration and LEDs description	5–10
Table 5–3 – Controller board configuration	5–12
Table 5–4 – Power supply board	5–13
Table 5–5 – Measurement board test points, LEDs description and connections	5–14
Table 5–6 – Photodiode board configuration	5–16
Table 5–7 – PM preamplifier board test points, LEDs description, connections and adjustments	5–17
Table 5–8 – UV lamp power board test points, LEDs description and connections	5–18
Table 5–9 – ARM20 board description	5–19

INDEX OF PAGES

Page	Date	Page	Date	Page	Date
0-1	02.2020	3-15	03.2019	5-1	03.2019
0-2	03.2019	3-16	03.2019	5-2	03.2019
0-3	03.2019	3-17	03.2019	5-3	03.2019
0-4	03.2019	3-18	03.2019	5-4	03.2019
0-5	03.2019	3-19	03.2019	5-5	03.2019
0-6	03.2019	3-20	03.2019	5-6	03.2019
0-7	03.2019	3-21	03.2019	5-7	03.2019
0-8	02.2020	3-22	03.2019	5-8	03.2019
		3-23	03.2019	5-9	03.2019
		3-24	03.2019	5-10	03.2019
1-1	03.2019	3-25	03.2019	5-11	03.2019
1-2	03.2019	3-26	03.2019	5-12	03.2019
1-3	03.2019	3-27	03.2019	5-13	03.2019
1-4	03.2019	3-28	03.2019	5-14	03.2019
1-5	03.2019	3-29	03.2019	5-15	03.2019
1-6	03.2019	3-30	03.2019	5-16	03.2019
1-7	03.2019	3-31	03.2019	5-17	03.2019
1-8	03.2019	3-32	03.2019	5-18	03.2019
1-9	02.2020	3-33	03.2019	5-19	03.2019
1-10	09.2019	3-34	03.2019	5-20	03.2019
1-11	03.2019	3-35	03.2019		
1-12	03.2019	3-36	03.2019		
		3-37	03.2019		
		3-38	03.2019		
		3-39	03.2019		
2-1	03.2019	3-40	03.2019		
2-2	03.2019	3-41	03.2019		
2-3	03.2019	3-42	03.2019		
2-4	03.2019	3-43	03.2019		
2-5	03.2019	3-44	03.2019		
2-6	03.2019	3-45	03.2019		
2-7	03.2019	3-46	03.2019		
2-8	03.2019				
2-9	03.2019				
2-10	03.2019				
2-11	03.2019	4-1	03.2019		
2-12	03.2019	4-2	03.2019		
2-13	03.2019	4-3	03.2019		
2-14	03.2019	4-4	03.2019		
		4-5	03.2019		
		4-6	03.2019		
3-1	03.2019	4-7	03.2019		
3-2	03.2019	4-8	03.2019		
3-3	03.2019	4-9	03.2019		
3-4	03.2019	4-10	03.2019		
3-5	03.2019	4-11	03.2019		
3-6	03.2019	4-12	03.2019		
3-7	03.2019	4-13	03.2019		
3-8	03.2019	4-14	03.2019		
3-9	03.2019	4-15	03.2019		
3-10	03.2019	4-16	03.2019		
3-11	03.2019	4-17	03.2019		
3-12	03.2019	4-18	03.2019		
3-13	03.2019				
3-14	03.2019				

CHAPTER 1

GENERAL - CHARACTERISTICS

1.1.	GENERAL	1-3
1.1.1.	INTRODUCTION	1-3
1.1.2.	DESCRIPTION	1-3
1.1.2.1.	Front panel	1-3
1.1.2.2.	Rear panel	1-4
1.1.2.3.	Internal view	1-6
1.1.3.	VARIOUS OPERATING MODES	1-8
1.1.3.1.	Standard	1-8
1.1.3.2.	Option	1-8
1.1.4.	ASSOCIATED EQUIPMENT (OPTIONAL)	1-8
1.2.	CHARACTERISTICS	1-9
1.2.1.	TECHNICAL CHARACTERISTICS	1-9
1.2.2.	STORAGE CHARACTERISTICS	1-10
1.2.3.	INSTALLATION CHARACTERISTICS	1-10
1.2.3.1.	Links between units	1-10
1.2.3.2.	Dimensions and weight	1-10
1.2.3.3.	Handling and storage	1-10
1.2.4.	STORAGE	1-10
	Figure 1-1 – ENVEA AF22e (with screen)	1-2
	Figure 1-2 – ENVEA AF22e* (without screen)	1-2
	Figure 1-3 – Color screen fitted with touch-screen	1-3
	Figure 1-4 – Rear panel	1-5
	Figure 1-5 – AF22e internal view	1-7
	Figure 1-6 – Links between units	1-10
	Figure 1-7 – AF22e dimensions (in mm)	1-11

1. GENERAL - CHARACTERISTICS



Figure 1-1 – ENVEA AF22e (with screen).



Figure 1-2 – ENVEA* (without screen).

1.1. GENERAL

1.1.1. INTRODUCTION

The AF22e is a continuous sulfur dioxide analyzer (with a detection limit of 0.4 ppb).

Its detection is based on ultraviolet fluorescence principle.

With the incorporation of recent optical and electronic technologies, the analyzer offers many advantages while requiring only limited maintenance.

The sample is taken using a Teflon tube (6 mm external diameter) connected to the analyzer rear panel. The sample is aspirated by an internal pump.

The measurement is displayed on a color display equipped with a touch screen located on the front panel.

This analyzer is also available in AF22e* (without screen) version.

1.1.2. DESCRIPTION

1.1.2.1. Front panel

The front panel includes the following:

- General switch.
- Backlit color TFT LCD (Thin Film Transistor Liquid Crystal Display):
 - Resolution of 800 x 480 (pixels), 7" screen,
 - The display provides the measurement values and other system parameters, as desired.
- Capacitive touch-screen projected on glass.



Figure 1-3 – Color screen fitted with touch-screen

1.1.2.3. Rear panel

All of the AF22e electrical connectors and gas inlets/outlets are located on the rear panel. Refer to Figure 1–4.

Gas inlets/outlets for standard analyzers (right-hand side):

- The inlet for the sample to be analyzed (1) is composed of 4 mm (I.D.) and 6 mm (O.D.) Teflon tube fittings, attached to the dust filter holder. The dust filter holder is equipped with a Teflon filtering diaphragm.
- The pump outlet (2), to exhaust the analyzed sample, consists of 4 mm (I.D.) and 6 mm (O.D.) Teflon fitting.
- A fan (9).
- The span gas inlet (3) is composed of 4 mm (I.D.) and 6 mm (O.D.) Teflon fitting to connect an external span gas at atmospheric pressure.

The following internal option is also located on the rear panel:

- The “zero” optional inlet (4) is composed of 4 mm (I.D.) and 6 mm (O.D.) pneumatic tube fitting. It is used to connect an external, SO₂-free, “zero air” supply at atmospheric pressure.

Gas inlets/outlets for analyzers with permeation bench option (right-hand side):

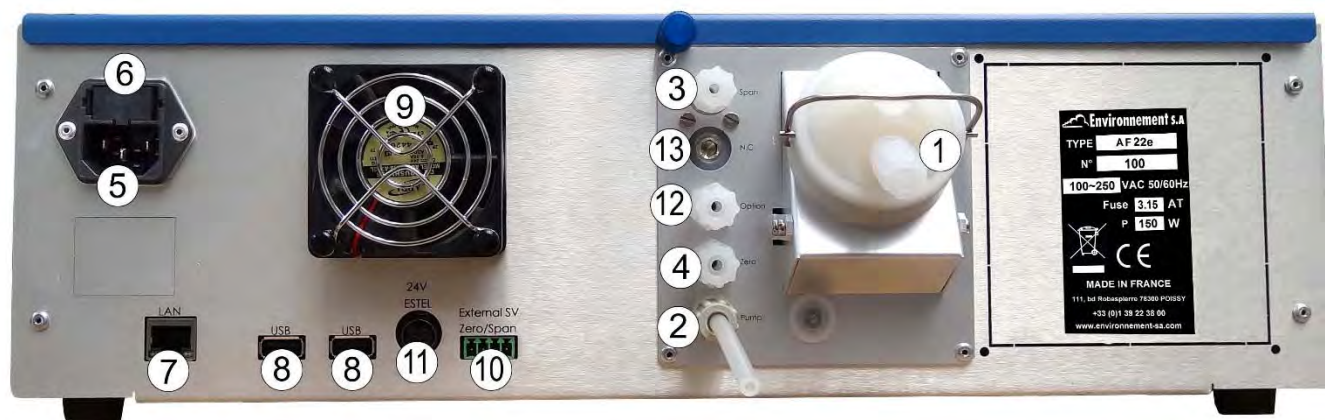
- The inlet for the sample to be analyzed (1) is composed of 4 mm (I.D.) and 6 mm (O.D.) Teflon tube fittings, attached to the dust filter holder. The dust filter holder is equipped with a Teflon filtering diaphragm.
- The pump outlet (2), to exhaust the analyzed sample, consists of 4 mm (I.D.) and 6 mm (O.D.) Teflon fitting.
- A fan (9).
- The span gas inlet (3) is stopped up.
- The zero air inlet for permeation bench (12).

The following internal options are also located on the rear panel:

- The “zero” optional inlet (4) is composed of 4 mm (I.D.) and 6 mm (O.D.) pneumatic tube fitting. It is used to connect an external, SO₂-free, “zero air” supply at atmospheric pressure.

Electrical equipment and connections for all analyzers (left-hand side):

- One main power supply block consisting of a three-contact socket (5) to connect a standard power cable and the general fuse (6): 3.15 A/250 V.
- One Ethernet output (7) and two USB ports (8).
- One 4-point BL connection to connect the optional solenoid valve for external calibration (10),
- One connection for the 24 V power supply for ESTEL board option (11).



(1) sample inlet, (2) pump outlet, (3) span inlet, (4) inlet for external zero air generator, (5) three-contact socket for main power supply, (6) general fuse, (7) Ethernet output, (8) two USB ports, (9) fan, (10) duplicate the zero and span solenoid valves for optional solenoid valve of external calibration, (11) 24 V power supply for ESTEL board option, (12) permeation bench inlet (optional), (13) not used.

Figure 1-4 – Rear panel

1.1.2.5. Internal view

The internal components of the analyzer are accessed by simply unscrewing the single knurled screw at the rear panel and sliding the upper cover.

The internal elements include (refer to Figure 1–5):

- Sample inlet dust filter mounted on the rear panel (1),
- Measurement module (2),
- Pump (3),
- ARM20 board (4),
- Interconnection board (5),
- 24 VDC power supply unit (6), with automatic universal input voltage 100~250 Vac 50/60Hz.
- Measurement board (7) that uses the + 24 V, + 12 V, + 5 V, +3.3 V power supplies. It supports analog signal acquisition and digital signal processing.
- Two pressure sensor boards (8),
- UV lamp supply board (9),
- Solenoid valve-filter assembly (13),
- An optional permeation bench block (not shown).

After passing through the dust filter (1), the sample to be analyzed passes through an hydrocarbon removing permeation tube, called the « carbon kicker ». The external carbon kicker tube is ventilated with under-vacuum zero air provided by the activated carbon filter.

The sample is then sent into the measurement module (2) located in a 50°C temperature-controlled enclosure.

This measurement module includes:

- The UV lamp (11) supplied by the UV lamp board (9). The Interconnection board (5) controls a chopper (12) allowing lamp beam interruption to measure the black photomultiplier current.
- A reaction chamber where fluorescence takes place.
- The detector unit consisting of a photomultiplier tube and a PM preamplifier board to measure the emitted SO₂ molecules fluorescence.
- The UV reference measurement unit consisting of one photodiode and the photodiode board. This set measures the UV lamp light intensity.

A diaphragm pump (3) aspirates the sample. A restrictor controls the sample flow.



(1) sample inlet dust filter, (2) measurement module, (3) pump, (4) ARM20 board, (5) Interconnection board, (6) 24 VDC power supply unit, (7) measurement board, (8) pressure sensor boards, (9) UV lamp supply board, (11) UV lamp, (12) shutter, (13) solenoid valve-filter assembly.

Figure 1-5 – AF22e internal view

1.1.3. VARIOUS OPERATING MODES**1.1.3.1. Standard**

- Programmable measurement range from 1 to 10 ppm, with a detection limit of 0.4 ppb for 36 second response time.
- Automated monitoring of parameters influencing metrology (UV emitted energy, gas flow, temperature and pressure) and correct operation tests.
- Measurement values in ppb or $\mu\text{g}/\text{m}^3$ (integrated conversion coefficient).
- Memory storage of average measurements in programmable periods.

1.1.3.2. Option

The following analyzer options are available:

- One ESTEL board case, as accessory, to be directly connected to USB ports on rear panel,
- One RS case delivered as an accessory, to be directly connected to USB ports on rear panel.
- A permeation bench.

1.1.4. ASSOCIATED EQUIPMENT (OPTIONAL)

- Analog recorders and data loggers.
- Digital data acquisition system.

1.3. CHARACTERISTICS

1.3.1. TECHNICAL CHARACTERISTICS

Measurement range :	– Programmable by user (0 to 10 ppm)
Units :	– ppb or $\mu\text{g}/\text{m}^3$ (programmable)
Noise (σ) :	– 0.2 ppb (response time: 36 seconds)
Minimum detectable limit (2σ) :	– 0.4 ppb (response time: 36 seconds)
Response time (0–90 %) :	– 20-120 seconds, fixed or automatic (programmable)
Zero drift :	– < 2 ppb / 24 hours in operation
Span drift :	– < 0.5 % / 24 hours
Linearity :	– $\pm 2\%$
H ₂ O influence :	– 2 ppb for 2 % H ₂ O
Temperature influence :	– 0.15 ppb / °C
Sample flow-rate :	– Approximately 0.333 liter/min
Display :	– TFT LCD color screen, resolution : 800 (RGB)x 480, size : 7 inches
Control :	– Touch screen
Output signals (option) :	– 4 analog outputs (0-1 V, 0-10 V, 0-20 mA, 4-20 mA)
Power supply :	– 100~250Vac, 50/60Hz + ground
Consumption :	– 280 VA when starting-up the unit – 110 VA in normal operation
Working temperature :	– + 10 °C to + 35 °C
Memory storage of measurement values :	– Capacity : 1 year – Flash EPROM storage
Alarm checks :	– Detection and indication of irregularities in continuous: temperature, flow rate, UV energy, too-high PM tube voltage, out of SO ₂ programmable measurement thresholds, calibration fault.
Maintenance test and diagnostics :	– Direct selection with the touch screen and/or remote selection with the embedded Web server.
Backup saving time for the real-time clock and RAM-stored data :	– 1 year maximum.
Ethernet output :	– RJ45 socket, UDP protocol.
USB port :	– Type A USB socket: 2.0 (3.0 compliant)
Zero/span external SV control :	– Contact connector with screw terminals

SV: Solenoid valve

1.3.3. STORAGE CHARACTERISTICS

Temperature: – 10 to 60°C.

1.3.4. INSTALLATION CHARACTERISTICS

1.3.4.1. Links between units

The AF22e analyzer uses the external links and power supplies illustrated in Figure 1–6 :

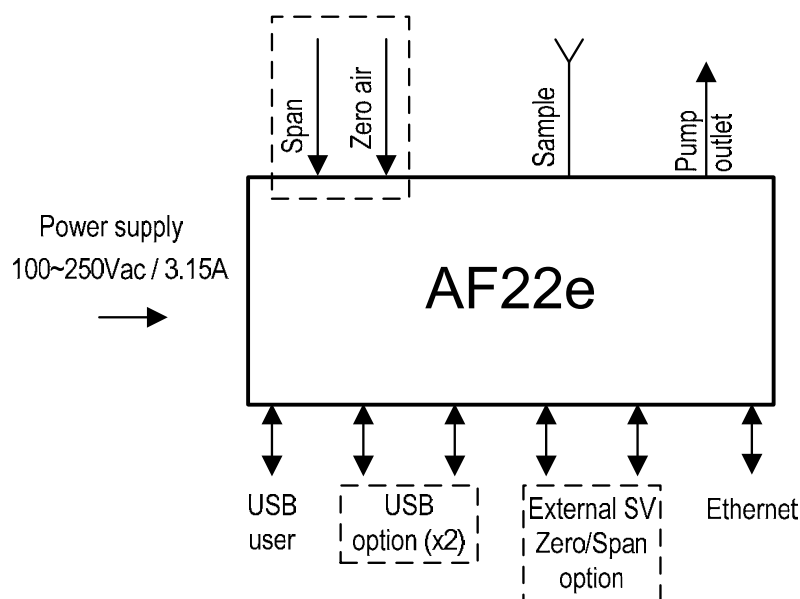


Figure 1–6 – Links between units

1.3.4.2. Dimensions and weight

The analyzer comes in a standard 19-inch, three-unit high rack.

Length : 606 mm

Width : 483 mm

Height : 133 mm

Weight : 9 kg

1.3.4.3. Handling and storage

The AF22e analyzer must be handled with care to avoid damage to the various connectors and fittings on the rear panel.

Make sure that the analyzer fluid inlets and outlets are protected with caps during handling.

1.3.5. STORAGE

The unit should be stored in the foam-packed case provided for this purpose.

Make sure that the analyzer fluid inlets and outlets are protected with caps during storage.

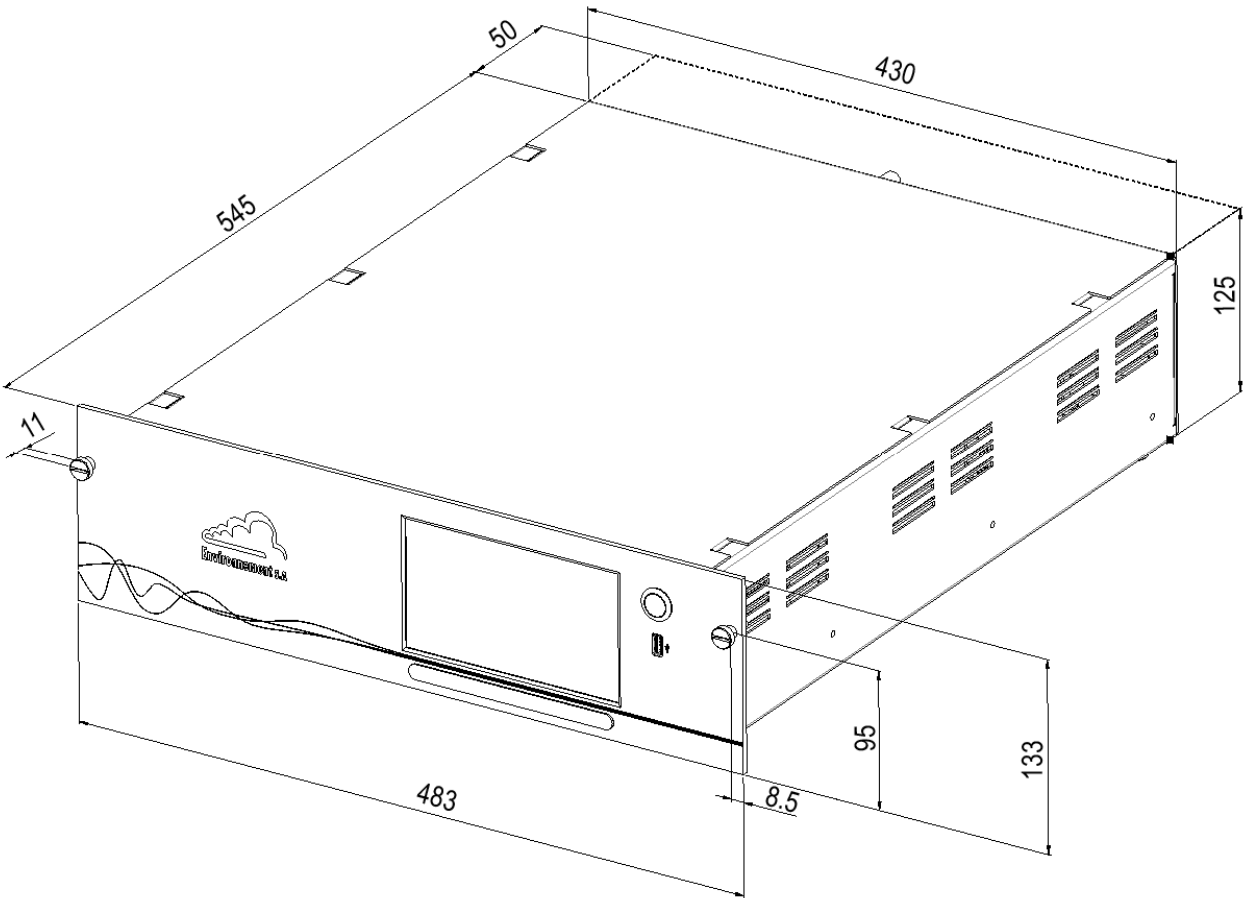


Figure 1-7 – AF22e dimensions (in mm)

Page intentionally left blank

CHAPTER 2

PRINCIPLE OF OPERATION

2.1	THEORETICAL BASIS	2-3
2.2	MEASUREMENT PRINCIPLE	2-6
2.3	GENERAL PRINCIPLE DIAGRAMS	2-7
2.4	SAMPLING AND ANALYSIS	2-9
2.5	SIMPLIFIED CALCULATION DESCRIPTION	2-11
2.6	PROGRAMMING THE RESPONSE TIME	2-12
2.7	ELECTRONIC ARCHITECTURE	2-13
2.8	NETWORK LINK AND USB PORTS	2-14

Figure 2-1 – Diagram showing molecule energy levels	2-5
Figure 2-2 – General principle diagram of standard AF22e	2-7
Figure 2-3 – General principle diagram of AF22e with permeation bench option	2-8
Figure 2-4 – Hydrocarbon molecule filtration	2-9
Figure 2-5 – Electronic architecture	2-14

Page intentionally left blank

2 PRINCIPLE OF OPERATION

2.1 THEORETICAL BASIS

Refer to Figure 2–1.

Energy diagram:

Many developments in quantum mechanics were performed in the early twentieth century. They allowed theoretical physicists to conceptualize processes involved in energy exchanges between a gaseous molecule such as sulfur dioxide and its environment.

The diagram shown in Figure 2–1 helps to easily understand the various phenomena involved in radiation absorption by the molecule.

Two essential comments clarify this diagram:

- Energy levels are quantized and distributed according to a structure which differs depending on the electronical, vibrational and rotational levels of the molecule.
- At the considered wavelengths, the "electron" energy scale includes only the energy levels of molecule valence electrons.

Likewise, transition from an electronical level to another by one-photon absorption always involves vibrational and rotational transitions of lower energy levels.

The mathematical formalism of molecular physics associates a time and spatial function named « wave function » with each molecule status. Then, a transition results in a new spatial « wave function » distribution.

Absorption and emission:

In its fundamental state E_{e0} , the SO_2 molecule can absorb only photons having sufficient energy to access the first excited state on the E_{e1} electronical scale.

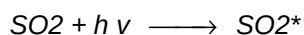
The photon energy E is given by Einstein's equation:

$$E = h\nu = h \frac{c}{\lambda}$$

Where :

- λ is the radiation wavelength output from a low-pressure zinc vapor lamp, given $\lambda = 213.9 \text{ nm}$,
- h is Plank's constant,
- c is vacuum light speed.

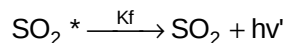
The transition may be written as:



The molecule always reaches a vibrational and rotational sub-level of E_{e1} (which is higher than E_{e1} level). From this E_{e1} sub-level, the molecule quickly scatters its vibration and rotation energy to return to E_{e1} level.

From E_{e1} , the molecule can return to any ground state sub level following different paths:

- by fluorescence:



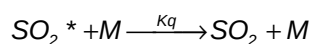
Based on the above formula,

⇒ Firstly, $\nu' = \frac{c}{\lambda'}$ is less than ν , therefore $\lambda' > \lambda$,

⇒ Secondly, λ' value follows a temperature dependent statistical distribution centered around an average wavelength.

The analyzer "monitors" photons emitted through a filter centered on 350 nm.

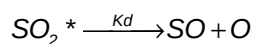
- by extinction



where M represents another gaseous molecule.

Then, the molecule scatters its energy mechanically and does not fluoresce. This phenomenon is known as « quenching ».

- by dissociation



The energy E_{e1} is sufficient to break the SO – O link.

K_f , K_d , K_q stand for quantum yields associated with each form of deactivation. Linked to the state E_{e1} lifetime, they express probabilities associated with each reaction type. They are determined by the whole space integral of the wave function product of the excited and fundamental states.

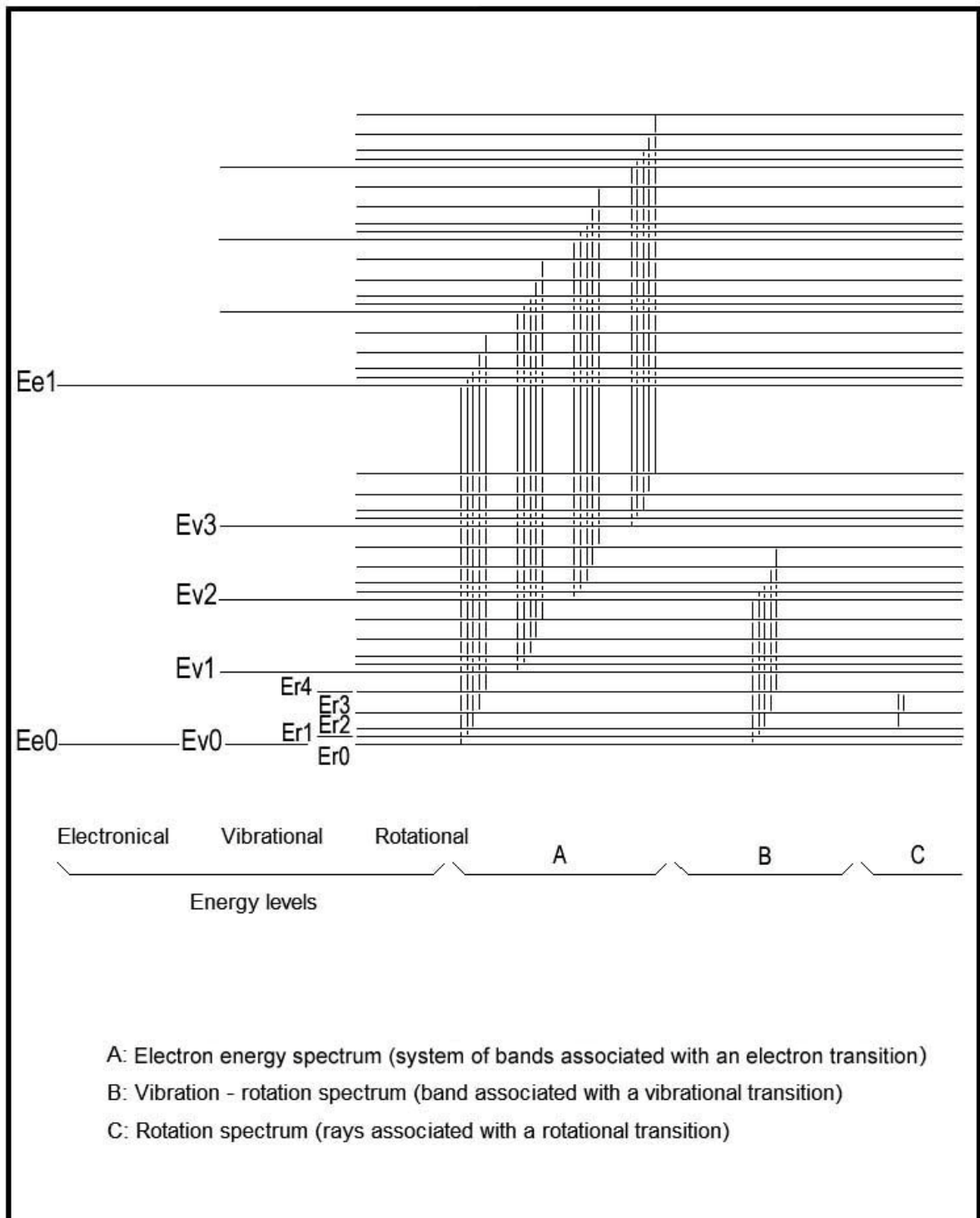


Figure 2-1 – Diagram showing molecule energy levels

2.2 MEASUREMENT PRINCIPLE

The radiation intensity absorbed by sulfur dioxide i_a in the L-length optical chamber follows the Beer-Lambert law:

$$i_a = i_0 \times (1 - e^{-\alpha LC})$$

where :

« i_0 » is the chamber inlet intensity,

« α » is the specific SO₂ absorption coefficient,

« C » is the gas concentration to be analyzed.

Likewise, the probability for an excited molecule to fluoresce is given by:

$$\frac{Kf}{Kf + Kq + Kd}$$

The fluorescence intensity i_f received by the PM is then expressed in the following form:

$$i_f = Gi_a \frac{Kf}{Kf + Kq + Kd}$$

where G is a constant depending on the lit chamber part seen by the PM.

Therefore:

$$i_f = Gi_0 \frac{Kf}{Kf + Kq + Kd} \times (1 - e^{-\alpha LC})$$

In our case, $\alpha L C \ll 1$ and $1 - e^{-\alpha LC}$ can be developed at first-order as follows:

$$1 - e^{-\alpha LC} \cong + \alpha LC$$

The result is:

$$i_f = \frac{Gi_0 Kf \alpha L}{Kf + Kq + Kd} C = \beta C$$

The PM senses radiation directly proportional to SO₂ concentration. This result is the basis of AF22e measurement technique.

2.3 GENERAL PRINCIPLE DIAGRAMS

They are presented in Figure 2–2 and Figure 2–3.

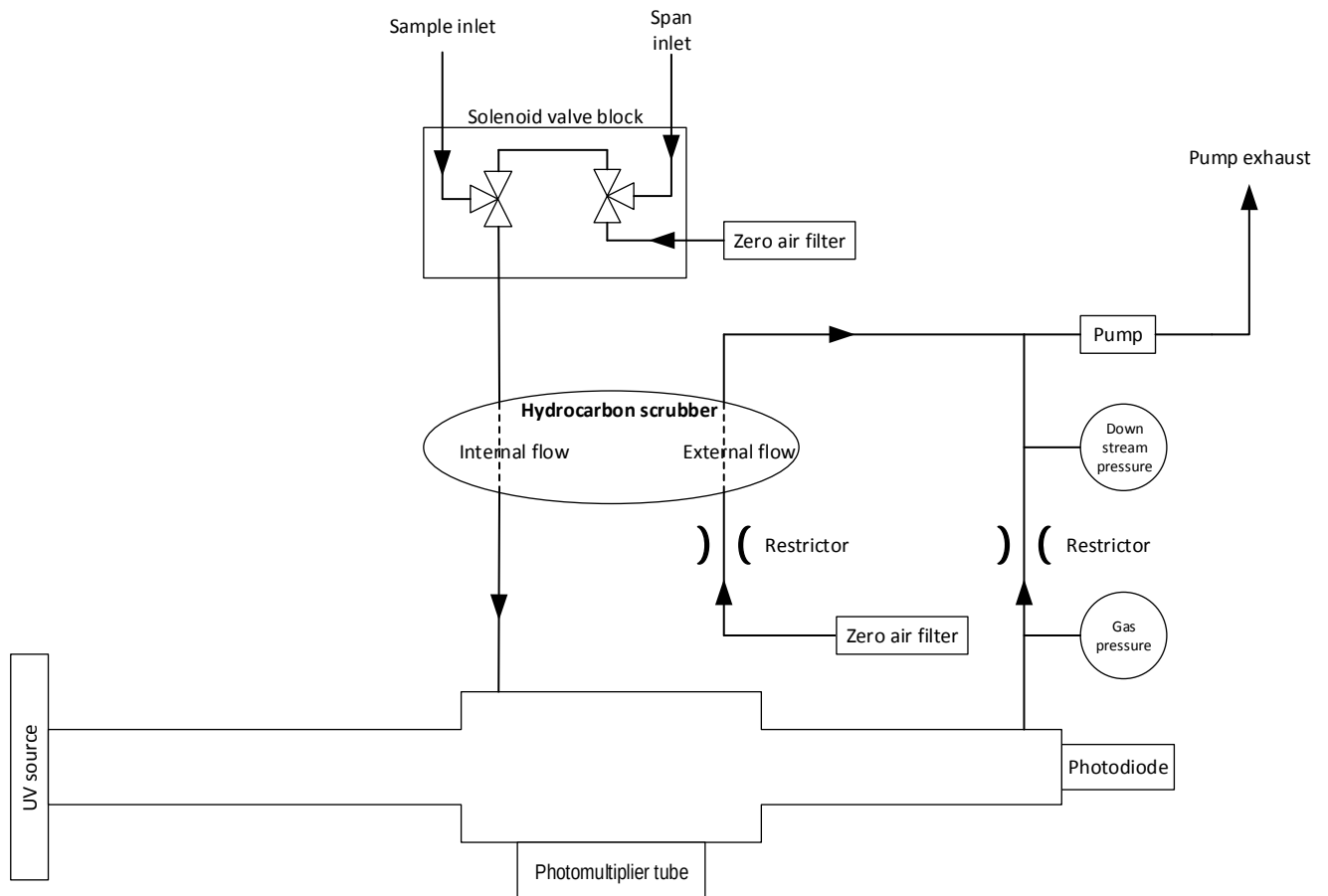


Figure 2–2 – General principle diagram of standard AF22e

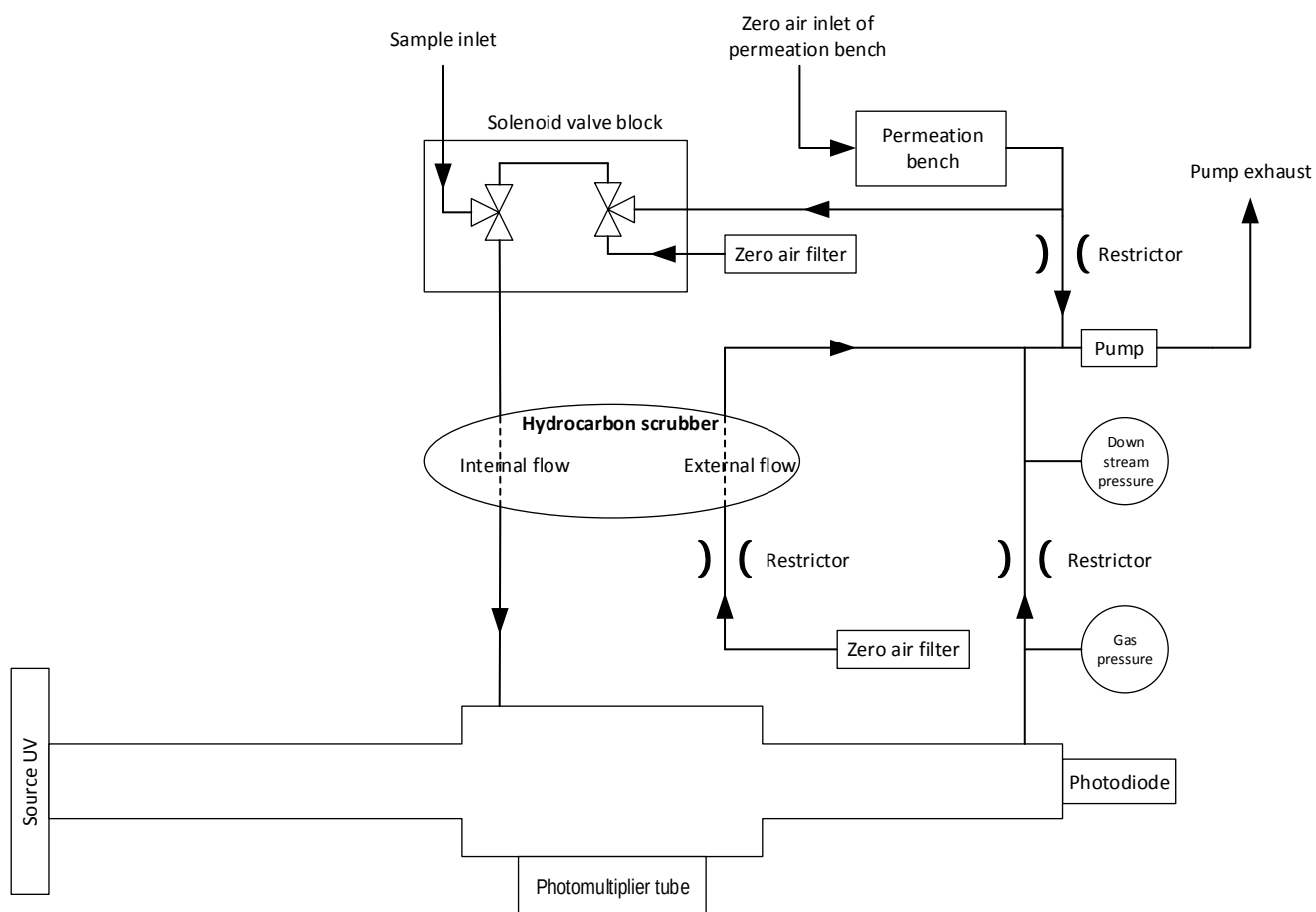


Figure 2-3 – General principle diagram of AF22e with permeation bench option

2.4 SAMPLING AND ANALYSIS

The sample is drawn through a Teflon tube with a pump connected to the analyzer rear panel. Dust protection is provided by the dust filter holder equipped with a Teflon filter diaphragm.

Hydrocarbon molecule filtration

Firstly, the sample to be analyzed is filtered by a HC removing aromatic molecule device. This device consists of two concentric tubes. The internal tube is made of silicone.

The sample containing HC aromatic molecules flows through the internal tube. The HC aromatic molecules are transferred by permeation to the external silicon tube. The transfer gradient is from higher to lower concentration.

The pump generates a vacuum in the external tube. The partial aromatic compound pressure decreases due to activated charcoal filtration. Molecules are exhausted outside the internal tube.

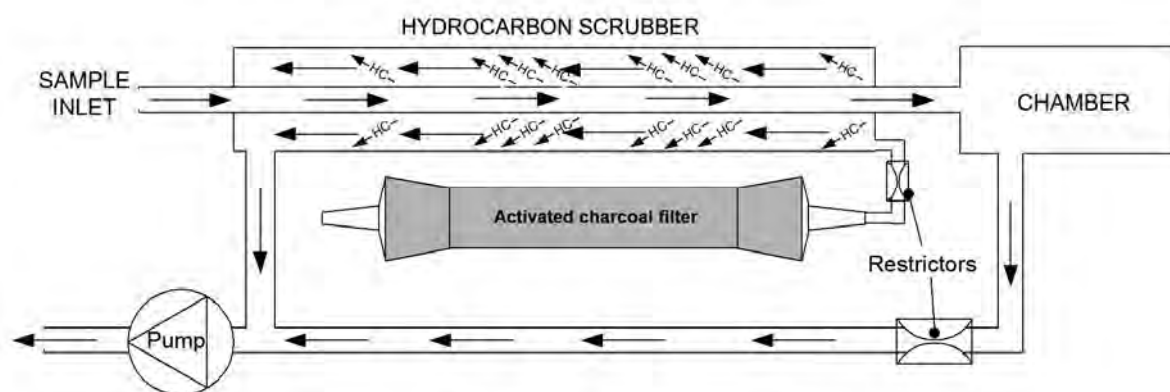


Figure 2-4 – Hydrocarbon molecule filtration

HC : hydrocarbon

Analysis

The HC-molecule free sample to be analyzed, is sent to a reaction chamber to be irradiated by an ultraviolet radiation centered at 214 nm, which is the SO₂ molecule absorption wavelength.

A photodiode measures the ultraviolet radiation generated by the UV lamp. This measurement is used during signal processing to compensate any variation of UV energy.

Molecules restore a specific fluorescence in the ultraviolet. The fluorescence is optically filtered between 300 and 400 nm to eliminate some interfering gases. The fluorescence is detected by the PM tube placed near the reaction chamber.

At each « zero-ref » start, a shutter is placed between the UV lamp and the reaction chamber inlet. This electrical zero corresponds to the PM tube darkness current and the preamplifier offset voltage. Taking account the electrical zero in signal processing removes any risk of temperature and over-time drift.

2.5 SIMPLIFIED CALCULATION DESCRIPTION

Calculations are performed following the steps below:

- During a zero-ref cycle, the analyzer in zero air measurement mode acquires the following signals:
 - Dark PM sig.: PM signal is recorded when shutter is closed,
 - PM Ref. sig.: PM signal is recorded when shutter is open,
 - Dark UV sig.: photodiode UV signal is recorded when shutter is closed,
 - UV ref. sig.: photodiode UV signal is recorded when shutter is open.

– Ratio calculation :
$$R_0 = \frac{PM\ ref - Dark\ PM\ sig.}{UV\ ref - Dark\ UV\ sig.} \quad (Equation\ 1)$$

- PM sig. (PM signal recorded on the ambient air) and UV sig. (UV photodiode signal recorded on ambient air) acquisition.

– Ratio calculation :
$$R = \frac{PM\ sig. - Dark\ PM\ sig.}{UV\ sig. - Dark\ UV\ sig.} \quad (Equation\ 2)$$

- $R_n, \dots, R_{n-3}$ are the R measurements at $t_n, \dots, t_{n-3}$ times.

- $Rmoy_n, \dots, Rmoy_{n-3}$ are calculated for each $t_n, \dots, t_{n-3}$ times by digital filtering of R_n ratio with the following equation :

$$Rmoy_n = \frac{1}{a_0} [(b_0 \times R_n + b_1 \times R_{n-1} + b_2 \times R_{n-2} + b_3 \times R_{n-3}) - (a_1 \times Rmoy_{n-1} + a_2 \times Rmoy_{n-2} + a_3 \times Rmoy_{n-3})] \quad (Equation\ 3)$$

Where:

$a_0, \dots, a_3$ and $b_0, \dots, b_3$ are determined by calculation,

- R_{zero} ratio is $Rmoy_n$ saved after a zero-reference

– Fluorescence calculation : $Mean = Rmoy - R_{zero} \quad (Equation\ 4)$

- Concentration calculation from linearization function :

$$Raw = F(Mean) \quad (Equation\ 5)$$

- Calibration, pressure and units correction application :

$$SO2 = Raw \times K_{etal} \times \frac{P_0}{P} \times K_{unit} \quad (Equation\ 6)$$

2.6 PROGRAMMING THE RESPONSE TIME

The automatic response time function may be activated or de-activated in the « *ADVANCED ANALYZER CONFIGURATION* » screen.

The minimum response time may also be modified in this menu.

The response time function changes the $a_0, \dots, a_3$ and $b_0, \dots, b_3$ coefficients according to the required response time used by the digital filter.

The desired response time is programmable in seconds from a choice list.

The response time can also be programmed in automatic mode, where by the analyzer automatically selects the most appropriate response time.

2.7 ELECTRONIC ARCHITECTURE

Communication between the ARM20 board and the other electronic boards (Measurement board, Interconnection board, Power supply board, Controller board, Pressure sensor board....) follows the schematic shown in Figure 2–5

The ARM20 board is a rapid calculation and interfacing (communication) board for the “e” series measurement modules. It is installed in all analyzers and offers one Ethernet output (RJ45 socket), and three USB ports (one on the front panel, and two on the rear panel through the Interconnection board). For these analyzers, the ARM20 board is the central element for electronic and metrological operation, and outward communication.

The measurement board performs measurement acquisition and operation parameter acquisitions. All metrological parameters are sent to the ARM20 board through a USB communication protocol.

The power supply board provides internal voltages from the 24 V cut-out power supply to the analyzer, as follows:

- +24 V filtered and secured,
- +12 V, for ARM20 board supplying,
- +5 V,
- +3.3 V.

The Controller board performs the main following functions:

- Provides (+ 24V) various power control for the internal solenoid valve switchings.
- Allows temperature acquisition and control management of the various internal heating elements.
- Allows pressure signal acquisition from the various pressure sensor boards.
- Integrates one USB hub to ensure communications between the various boards.

The Interconnection board allows connections of all the connectors and sensors together.



Three USB ports are available in the “e” series electronic architecture.

CHAPTER 3

OPERATING INSTRUCTIONS

3.1.	INITIAL STARTUP	3-3
3.1.1.	PRELIMINARY OPERATIONS	3-3
3.1.2.	STARTING UP THE UNIT	3-5
3.2.	PROGRAMMING THE AF22E	3-9
3.2.1.	SCREEN AND KEYBORAD DESCRIPTION AND USE INSTRUCTIONS	3-9
3.2.1.1.	Definition of the control screen areas	3-9
3.2.1.2.	Contextual help functioning	3-10
3.2.1.3.	Virtual keyboard input	3-10
3.2.2.	OPERATION PARAMETER PROGRAMMING	3-12
3.2.2.1.	Field requiring character input	3-12
3.2.2.2.	Fields requiring parameter selection in a list	3-12
3.2.2.3.	OFF/ON status modification button	3-12
3.3.	BROWSING ERGONOMY	3-13
3.4.	ANALYZER FUNCTION DESCRIPTION	3-18
3.4.1.	STANDARD FUNCTIONS	3-18
3.4.1.1.	Analyzer controls	3-18
3.4.1.2.	Measurement channel display	3-20
3.4.1.3.	Measurement channel curve display	3-22
3.4.1.4.	Analyzer synoptic diagram	3-25
3.4.2.	ADVANCED FUNCTIONS	3-26
3.4.2.1.	Password management	3-26
3.4.2.2.	General configuration	3-27
3.4.2.3.	Automatic cycle configuration	3-28
3.4.2.4.	Language selection	3-29
3.4.2.5.	Find and display recorded averages	3-29
3.4.2.6.	Delete all recorded averages	3-31
3.4.2.7.	List of connected clients	3-31
3.4.2.8.	Diagnostic functions	3-31
3.4.2.9.	Set the analyzer in standby mode	3-34
3.4.2.10.	Set the analyzer in maintenance mode	3-34
3.4.2.11.	Advanced analyzer configuration	3-35

3.4.2.12. Automatic linearization of the flow	3–36
3.4.2.13. Airproof test	3–37
3.4.2.14. Potentiometer setting	3–37
3.5. CALIBRATION	3–38
3.5.1. OVERVIEW OF CALIBRATION AND CONCEPTS	3–38
3.5.1.1. Zero air generation	3–39
3.5.1.2. Span gas generation (SO ₂)	3–40
3.5.1.3. Internal solenoid valves	3–40
3.5.2. ZERO AND SPAN POINT CHECK	3–41
3.5.2.1. Equipment Required	3–41
3.5.2.2. Procedure	3–41
3.5.2.3. Use of automatic cycles	3–41
3.5.3. TWO-POINT CALIBRATION	3–43
3.5.3.1. Equipment required	3–43
3.5.3.2. Procedure	3–43
3.5.4. MULTI-POINT CALIBRATION	3–44
3.5.4.1. Overview	3–44
3.5.4.2. Equipment Required	3–44
3.5.4.3. Procedure	3–45
 Figure 3–1 – Fluid and electrical connections	 3–3
Figure 3–2 – Installation of the sample gas inlet	3–4
Figure 3–3 – Example of a pressurized gas connection	3–39
Figure 3–4 – Typical calibration device diagram	3–44

3. OPERATING INSTRUCTIONS

3.1. INITIAL STARTUP

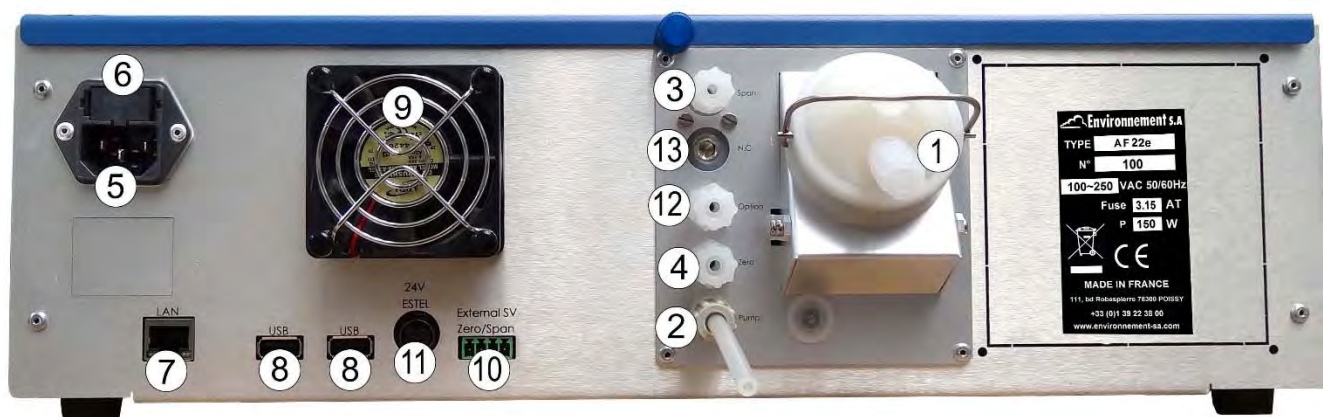
The analyzer is inspected before delivery. Calibration is verified at the factory.

3.1.1. PRELIMINARY OPERATIONS

Refer to Figure 3–1.

Start-up involves the following preliminary operations:

- Visually examine the inside of the instrument to make sure no damage occurred during transport.
- Remove the caps from the gas inlet and outlet (1) (2) on the unit (set aside for future storage).
- Make sure a Teflon filter diaphragm is inside the dust filter, and then connect the 4 mm (I.D.) and 6 mm (O.D.) air sampling tube to the sample inlet (1).
- Connect the Ethernet output (7).
- If necessary, connect accessories via USB ports (8),
- Connect the main power supply cable to a socket 230 V, 50 Hz + ground or 115 V, 60 Hz + ground.
- Connect the pump outlet (2) to the vent.



(1) sample inlet, (2) pump outlet, (3) span inlet, (4) inlet for external zero air generator, (5) three-contact socket for main power supply, (6) general fuse, (7) Ethernet output, (8) two USB ports, (9) fan, (10) duplicate the zero and span solenoid valves for optional solenoid valve of external calibration, (11) 24 V power supply for ESTEL board option, (12) permeation bench inlet (optional), (13) not used.

Figure 3–1 – Fluid and electrical connections

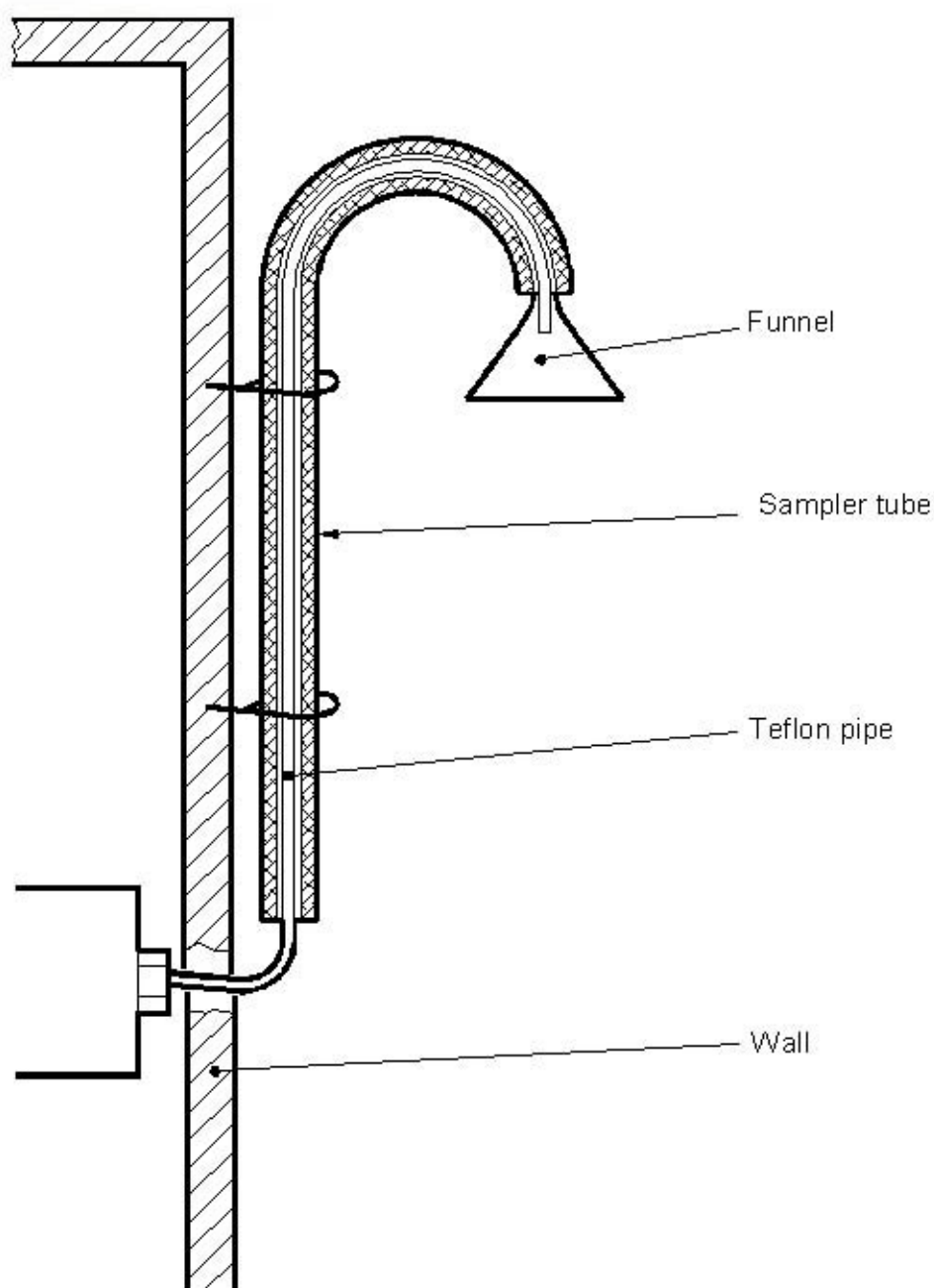


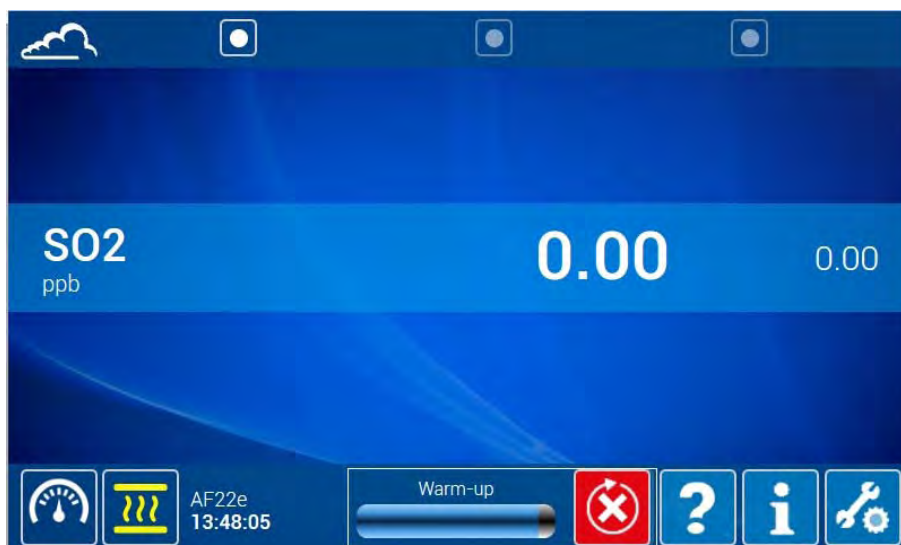
Figure 3-2 – Installation of the sample gas inlet

NOTE : Recommended height for sampling tap: 2.50 m
Maximum recommended length of Teflon sampling pipe: 6 m.

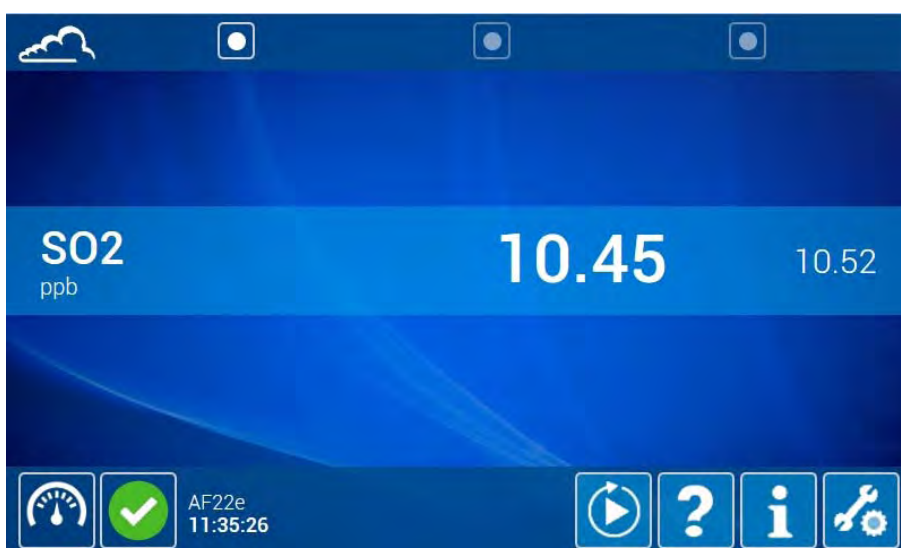
3.1.2. STARTING UP THE UNIT

Starting up the unit consists in carrying out, in the following order, the INSTALLATION / COMMISSIONING procedure described here-below.

1/ Press the ON/OFF push button located on the front panel. The analyzer starts running, it passes into warm-up cycle (duration of which is 1800 seconds maximum). The homepage is displayed: it allows to visualize the warm-up progress period.



The warm-up period is ended when all the metrology parameters are within the operational limits. The warm-up progress icon will disappear and the following homepage is displayed:



This screen is the analyzer standard function homepage. It contains additional pages that are displayed by activating the buttons  to , present at the top of the screen. These pages allow the measurement channel display, the « *Real-time graph* » of the instantaneous measurement channel values, and the « *Analyzer synoptic* » giving the main parameter values.

By using the touch screen mounted on the analyzer front panel: no password is required to activate the standard functions.

By using a remote PC: a password is required to be input in the pop-up that will appear. Enter the User password given in the « **Warning** » section of the Quick-Start, and validate with .

Button and icon definitions specific to these screens

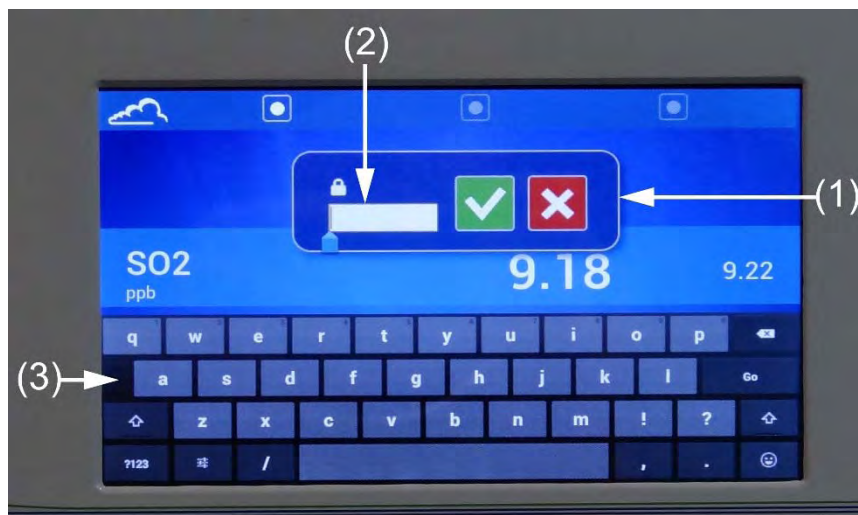
	Displays the selection buttons of the analyzer fluid inlets:  sample inlet,  zero inlet,  span inlet.
	Warm-up status icon: indicates that analyzer is warming-up.
	Normal operation status icon: indicates that the analyzer operates normally.
	Information area: indicates the analyzer model and time.
	Allows to visualize the warm-up progress status
	Stop button of the current cycle
	Displays the starting buttons of the analyzer cycles:  starts calibration cycle,  starts zero reference cycle.
	Displays the contextual help
	Displays analyzer information panel.
	Gives access to the advanced analyzer functionalities.

2/ Access to the advanced analyzer functionalities

By using the touch screen mounted on the analyzer front panel: touch the button  in order to open the User password input pop-up (1). Touch the field (2) to display the QWERTY keyboard (3).

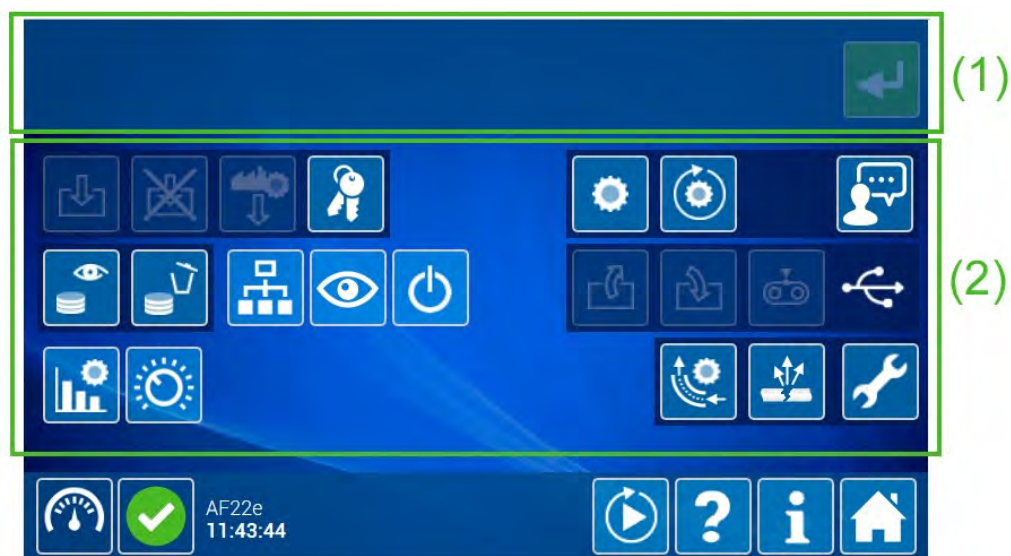
Touch  in order to switch alphanumeric keyboard to numeric keyboard. Enter the User password and validate with : the pop-up and the keyboard closes, and the advanced functionalities homepage opens.

By using a remote PC: do the same with the PC keyboard and mouse.



The button  allows to close the pop-up without input validation.

The home page of the advanced analyzer functionalities is presented below:



The accessible functions for the User appear in white with a blue background in zone (2). The function access to the functions depends of the password's level that was entered.

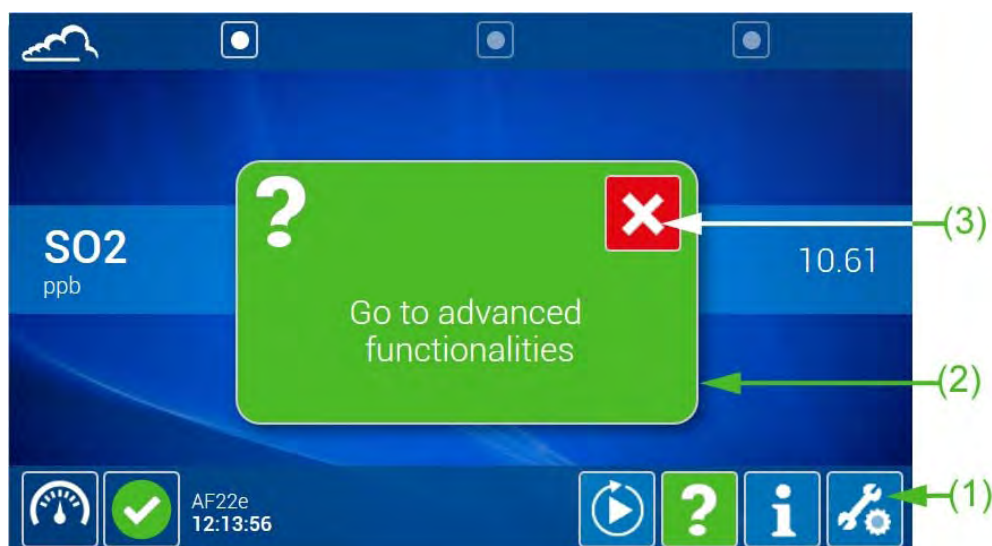
Each accessible function is activated by contact, it then changes color to green. The name of the selected function will appear in zone (1), and the button  is highlighted . By double clicking on the function's icon or by using the button , the user can access to the corresponding screen.

3/ Contextual help functioning

The contextual help is directly accessible through all the screens. The user activates it by touching / clicking on the button  that will become green . Then, the user touches / clicks on the buttons or on a parametric screen part in order to display a pop-up that describes the use of the selected function or button. The pop-up is closed by clicking on .

It is necessary to touch/ click on  again to deactivate the contextual help.

See the example below:



(1) Activated button (highlighted), (2) pop-up message explaining the function of the activated button, (3) icon to be used in order to close the pop-up.

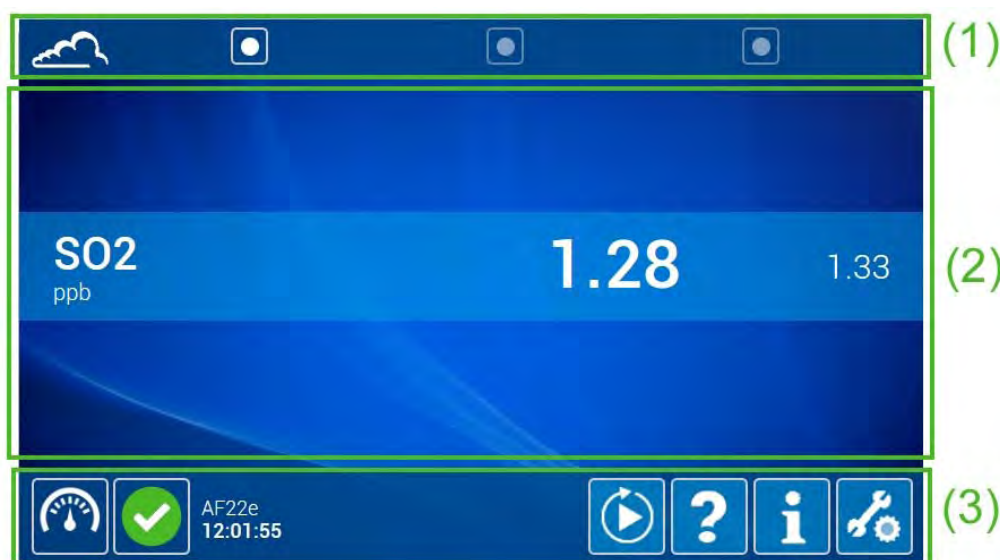
3.2. PROGRAMMING THE AF22E

3.2.1. SCREEN AND KEYBORAD DESCRIPTION AND USE INSTRUCTIONS

The touch screen mounted on the front panel is alternately used to visualize the control screens and the virtual input screen allowing input and modification of the analyzer parameters. The control button activation is done by touching.

When the analyzer is connected to a remote computer, the parameters are modified using the computer keyboard, and the control buttons are activated by clicking with the mouse.

3.2.1.1. Definition of the control screen areas



The areas of control screens are defined as follow:

(1)	Browsing area proper to the current screen: when the screen has more than one page, activate the button to allow to display the corresponding page.
(2)	Measurement or configuration area. It displays the measurement parameters (gas, value, units...) or the configurable parameters associated with the selected menu.
(3)	Manual control area: inlet fluid selection and manual launching of calibration and zero reference cycles. Information area and advanced functionality access.

Control and information area (3) description

This area is present in all the screens.

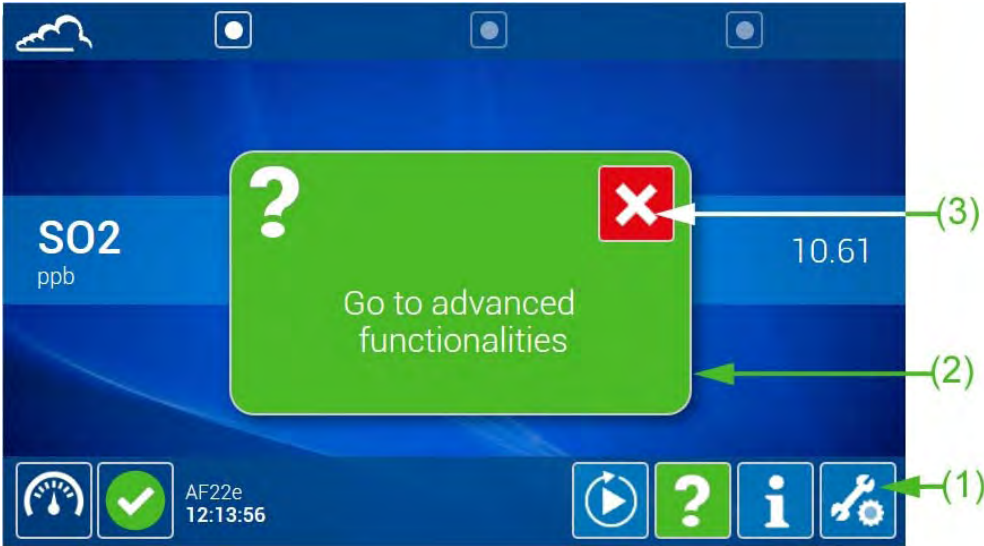
	Activating this button displays the buttons allowing to manually select the fluid inlets of the analyzer: sample inlet, zero inlet, span inlet.
	The icon displayed in this place indicates the currently operating status of the analyzer. The various status icons are: normal operation, warm-up, control, alarm, network disconnected, maintenance, stand-up.
	Information area indicating model and current time of the analyzer.
	Activating this button displays the buttons allowing to manually start the analyzer cycles: starts calibration cycle, starts zero reference cycle.

	Activating this button displays the contextual help.
	Activating this button displays the analyzer information panel.
	Activating this button gives access to the advanced analyzer functionality home page.

3.2.1.2. Contextual help functioning

The contextual help is directly accessible through all the screens. The user activates it by touching / clicking on the button  that will become green . Then, the user touches / clicks on the buttons or on a parametric screen part in order to display a pop-up that describes the use of the selected function or button. The pop-up is closed by clicking on .

See the example below:



(1) Activated button (highlighted), (2) pop-up message explaining the function of the activated button, (3) icon to be used in order to close the pop-up.

3.2.1.3. Virtual keyboard input

The virtual input screen is used to modify the control screen parameters displayed from the front panel touchscreen. This screen is displayed instantly in the lower half-part of the screen when the user touches an input field to be modified. Two touch-button keyboard types are available: numeric and alphanumeric QWERTY. They are displayed depending on the input field nature: numeric keyboard if the field is numeric type, alphanumeric keyboard if the field requires number and letter inputs.

The examples below are showing the keyboard display in the « *General configuration* » screen. The keyboard operation is identical for all screens.

When the user presses down a key, the button (1)  at the top right corner is activated and highlighted to . It is used to validate the whole inputs done in the screen, and to close the screen after.

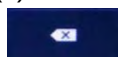
NOTE : Field inputs are indexed.
These keyboard keys work like Android system keys found on tablets and smart-phones.

The numeric keyboard:

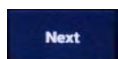
It is displayed in the area (2) of the screen below:



(1) button for input validation, (2) numeric keyboard with touchscreens.



: clears digits during input.



: used to move from an input field to another according to its indexation.

– The alphanumeric keyboard:



(1) button for input validation, (2) alphanumeric keyboard with touchscreens.



: switches from alphanumeric to numeric keyboard.



: switches from numeric to alphanumeric keyboard.



: switches from numeric to symbol keyboard.



: validates input of the considered field.

3.2.2. OPERATION PARAMETER PROGRAMMING

The standard functionalities cannot be modified. The user can only modify the advanced functionality parameters.

3.2.2.1. Field requiring character input

- Using the front panel touchscreen :

Touch the input field to be modified: if the field is alphanumeric, the QWERTY keyboard is displayed, if the field is numeric, the numeric keyboard is displayed.

Input with the touch keys, and validate with . The keyboard closes and the input field is displayed properly filled in the control screen.

- From a remote PC:

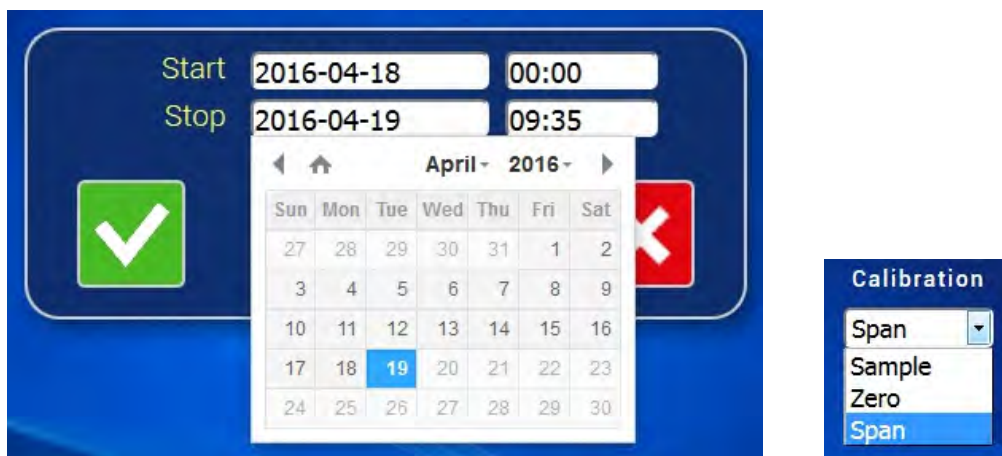
Use keyboard and mouse of remote PC to input, modify and validate the field characters.

3.2.2.2. Fields requiring parameter selection in a list

Many cases are possible, depending on the input type.

Touch the field to be modified. The parameter or value list to be selected is displayed: the current parameter or value is displayed white in blue background. Touch (or click on with the remote computer mouse) the new parameter or value to be selected. The field is displayed white in blue background.

Validate with . The list closes, and this new parameter or this new value replaces the previous.



3.2.2.3. OFF/ON status modification button

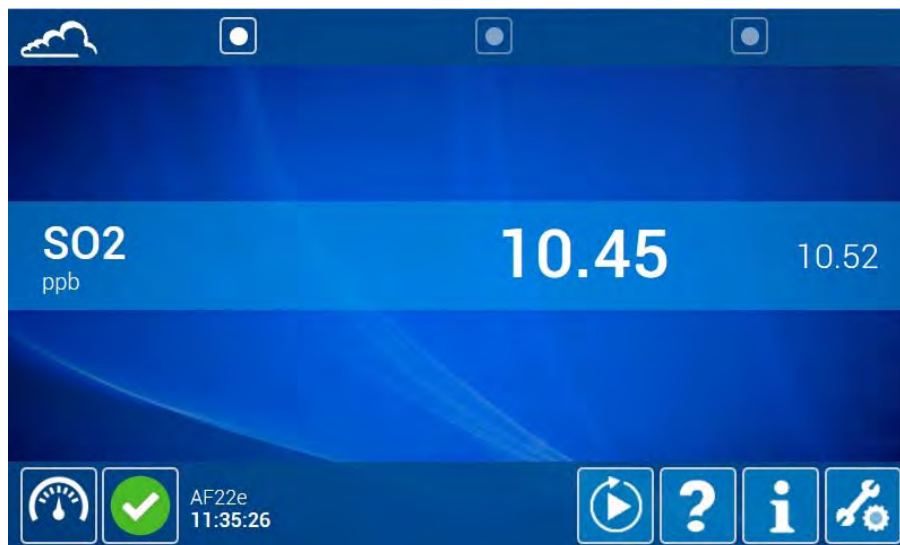
To switch OFF to ON, touch (or click on) the white button. The ON field becomes white in blue background. And vice versa.



NOTE : When many modifications are performed in the same screen, the user must ALWAYS validate with the key  to take into account the whole modifications.

3.3. BROWSING ERGONOMY

The AF22e home page is the following:

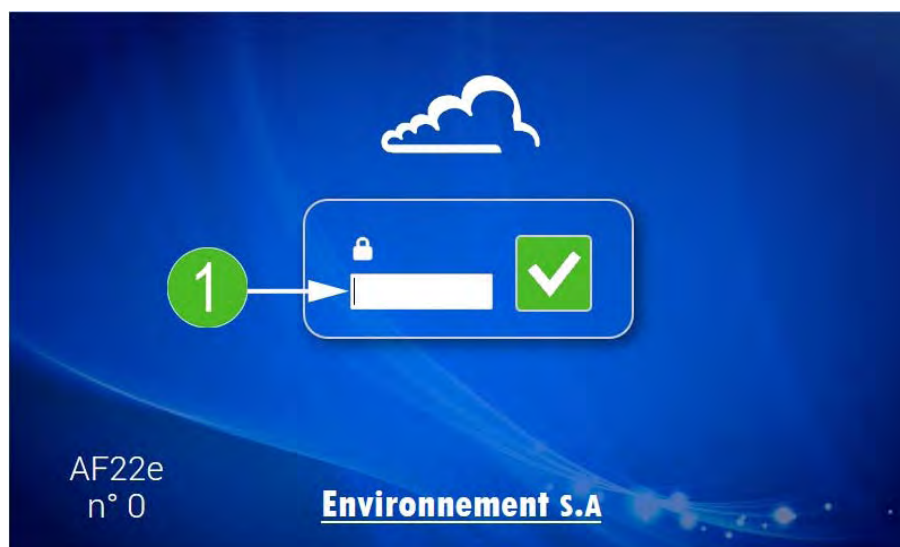


By default, the home page displays the SO₂ measurement channel.

From the front panel touchscreen, the homepage allows free (i.e. without password) and direct (i.e. the first browsing level) access to the standard functionalities of the analyzer.

Activating the buttons of top screen  to  allows to drag display sideways to view the « *Real-time graph* » of the instantaneous values of the selected measurement channels, and the « *Analyzer synoptic* » indicating the main operation parameter values.

From a remote PC, the page below is displayed first:

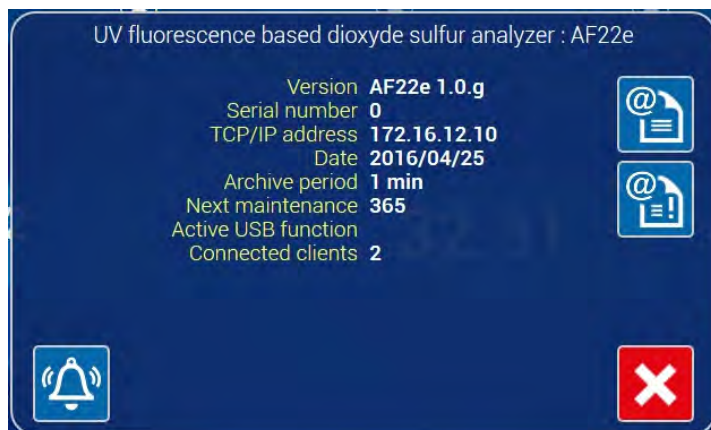


The user enters the password in the field (1) and validates with  to open the home page.

Standard functionalities are the followings:

	Display of fluid inlet selection buttons.
	Manual selection of sample inlet.
	Manual selection of zero inlet.
	Manual selection of span inlet.
	Display of analyzer status.
	Indication of analyzer model and time.
	Display of analyzer cycle launching buttons.
	Manual start of calibration cycle.
	Manual start of zero reference cycle.
	Display of contextual help.
	Display of analyzer information panel.
	Access to the advanced analyzer functionalities.
	Access to homepage.

The information panel displays the software version, analyzer serial number, TCP/IP analyzer address, current date programmed in the analyzer, memorization period for data archiving, remaining days up to next maintenance, if YES or NO an USB function is active on the analyzer, the number of connected clients.



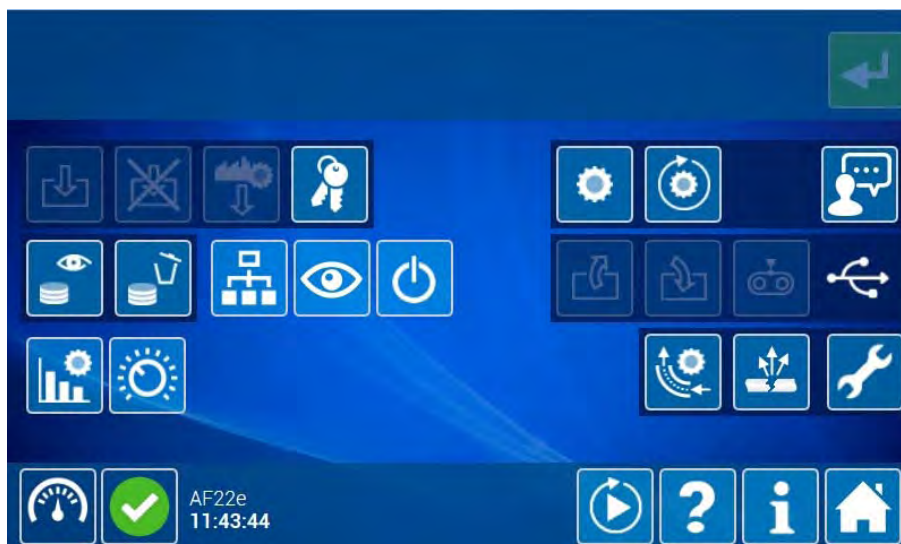
Icon and button definition specific to the information panel

	Means that the analyzer is equipped with a WIFI key for remote connection from a tablet or a smartphone. This icon goes with WIFI TCP/IP identifier and address required for connection.
	Allows to email analyzer status (mux signal values).
	Allows to email recorded events (history)
	Sound signal to locate the corresponding analyzer when not fitted with screen.
	Closes the information panel.

The advanced function access is selectively assigned to password bearers only. There are three possible authorization levels for passwords: user level, advanced level, expert level.

The user-accessible functions are white in blue background. They depend on the hierarchical level of the assigned password.

The access page to the advanced functions is the following:



The analyzer advanced functions are the followings:

	Restore factory settings. It is recommended to backup before executing this operation
	Passwords management
	General configuration (Address, Protocol, Measurement channels...)
	Automatic cycles configuration
	Language selection
	Find and display recorded averages
	Delete all recorded averages (Warning ! this operation can't be cancelled)
	List of connected clients
	Diagnostic functions (alarm, input / output, mux...)
	Set the analyzer in standby mode
	Advanced analyzer configuration
	Automatic linearization of flow
	Test for leaks

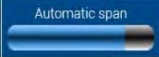
	Set the analyzer in maintenance mode
	Return to standard functionality home page
	Give the USB characteristics (free and total memory) when it is connected to the analyzer.
	Backup of configuration and software on USB when it is connected to the analyzer.
	Restore software and configuration from USB when it is connected to the analyzer
	Record instantaneous measurement on USB when it is connected to the analyzer
	Check the potentiometer setting of the PM amplifier board.

3.4. ANALYZER FUNCTION DESCRIPTION

3.4.1. STANDARD FUNCTIONS

3.4.1.1. Analyzer controls

The analyzer controls are the followings:

Controls	Function description
 Selects sample inlet	This function switches the analyzer to the sample gas inlet. The gas is continuously sampled through the dust inlet filter. This mode can be interrupted at any time by manually selecting another gas inlet (zero or span). The unit is selected in the « <i>Advanced analyzer configuration</i> » screen.
 Selects zero inlet	This function switches the analyzer to the zero gas inlet. This operation allows stability checking and zero drift to determine the need to launch a zero reference cycle or program its repetition period.
 Selects span gas inlet	This function switches the analyzer to the span inlet. The measured span gas value, possibly increased by a programmed offset, is displayed on the screen. This operation allows checking the stability and the span drift to determine the necessity of launching an auto-span cycle or programming its repetition period. An external calibrated span must be connected to this inlet.
 Launches a calibration cycle	This function launches manually an automatic span cycle. The user activates  and views cycle progression in the scroll bar . At any time, it is possible to abort the current cycle by touching , or stop and validating the current cycle by touching . The current cycle stop results in « Span Coeff. » value modification. During an automatic calibration cycle, the analyzer adjusts its « Span Coeff.» automatically to equal its reading value (minus the programmed offset) with span gas concentration. The span gas concentrations are programmable in the « <i>Advanced analyzer configuration</i> » screen. Adjustment of the cycle duration takes place in the « <i>Duration</i> » field of the « <i>Automatic cycle configuration</i> » screen. The cycle is finished when the scroll bar is completely grey, , it closes automatically.



Launches a zero reference cycle

This function launches manually an automatic zero correction cycle to adjust the zero drift.

The user activates  and visualizes cycle progression in the scroll bar



At any time, it is possible to abort the current cycle by touching , or stop and validating the current cycle by touching .

The cycle is finished when the scroll bar is completely grey, , it closes automatically.

3.4.1.2. Measurement channel display

By default, the homepage displays the compound channel to be measured:



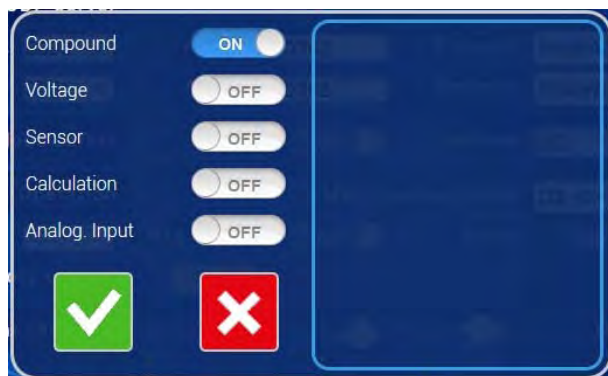
To display additional measurement channels, it is first necessary to select them in the « *Measurement channel* » section of the « *General configuration* » screen shown below (see outlined frame with arrow):



Touch button definitions specific to this section:

	Deletes the selected measurement channel
	Moves upwards the selected measurement channel
	Moves downwards the selected measurement channel
	Adds a measurement channel

To select additional channels, activate and open the selection pop-up. There are five channel groups: compound, voltage, sensor, calculation, analog input.



Then, to activate  to  in order to display the corresponding group elements, touch/click on the elements to be selected : they become sky-blue highlighted.

Once they are selected, the channels no more appear in the list. This allows not to select the same channel twice.

Then, activates  to validate this selection and close the pop-up. Do the same for the other channel groups: it is possible to select the whole available channels. The selected measurement channels take place in the numbered fields (see outlined frame with arrow):



When selection is finished, activates  to definitively validate the whole selection, then activates  to display the measurement channels in the home screen (home page):



Three measurement channels are displayed per page. To display the whole selected channels, the GUI (graphic user interface) adds as many pages as necessary. These additional pages are symbolized by the buttons  and  displayed at the screen top (see the outlined frame with arrow). The user clicks-on/touches these buttons to display pages successively.

3.4.1.3. Measurement channel curve display

The user views the measurement channel graphs by touching / clicking-on the next-to-last button in the browsing bar at screen top (see outlined frame with arrow). If no channel was previously selected, the following message is displayed:



Touch / click-on the channel name to select: they change color and are displayed in green, as shown in the screen below:



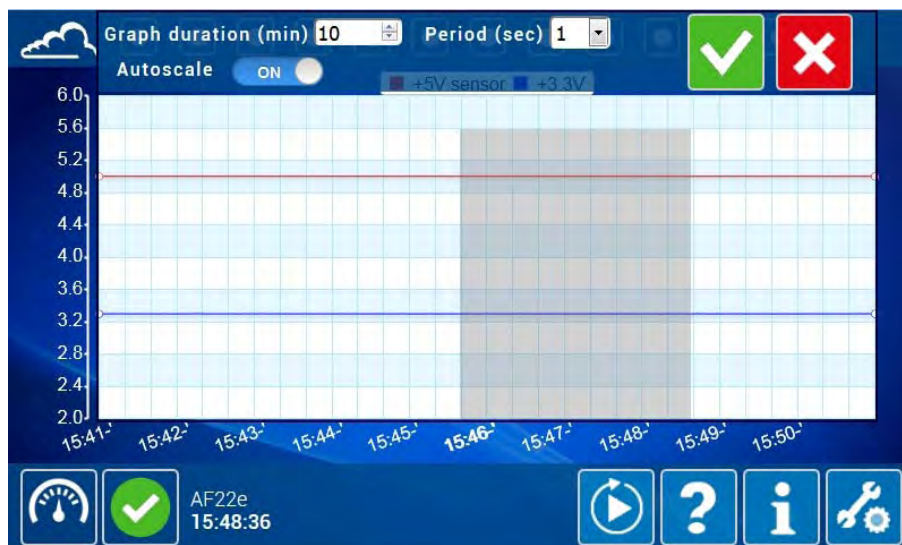
Go back to the next-to-last page to display the corresponding curves:



The Y-axis scale setting is automatic and adjusted in real time for current display optimization. Maximum six curves can be displayed simultaneously. By touching / clicking in the graph, the user activates the zoom function with the manual scale, as shown in the screen below:



The user selects the screen zone he wants to zoom:



Then, he activates  to validate: the selected zone is zoomed-in in the screen.



To exit zoom function, the user clicks in the screen two times, then he activates the action with 

This zoom function is available for X-axis and Y-axis.

Moreover, the user can deactivate the auto scale and he can set the desired Y-axis scale.

3.4.1.4. Analyzer synoptic diagram

The user views the synoptic diagram by clicking-on / touching the last button in the browsing bar at screen top (see outlined frame with arrow). This screen displays the entire fluid circuit and the significant operating parameter values.

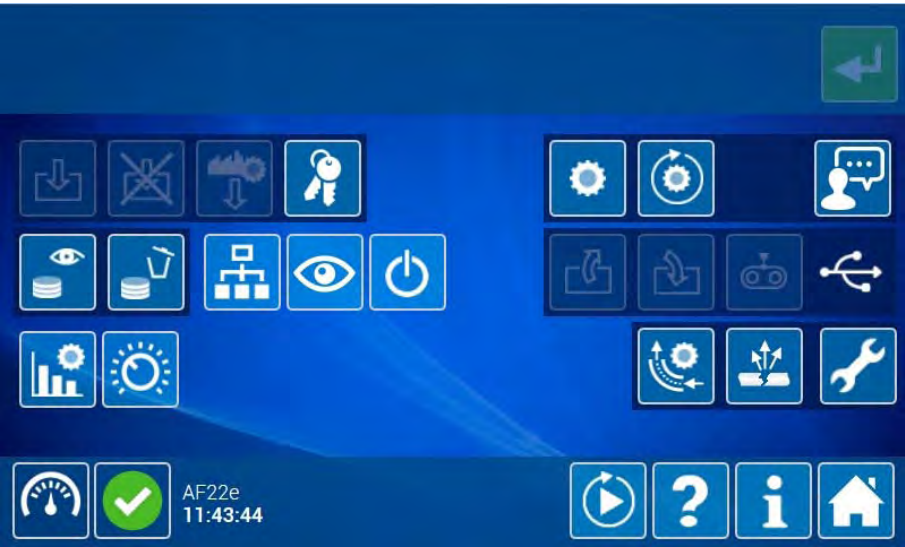


Screen legends are as follows:

- (1): SO₂ measurement
- (2): internal temperature of the analyzer,
- (3): photodiode UV signal,
- (4): UV lamp intensity,
- (5): photo multiplier tube high voltage,
- (6): photo multiplier tube signal,
- (7): chamber pressure,
- (8): chamber temperature,
- (9): sample flow rate,
- (10): zero air filter autonomy,
- (11): filter autonomy of carbon kicker.

3.4.2. ADVANCED FUNCTIONS

The user accesses the advanced functions homepage by touching / clicking-on .



The advanced functions can only be accessed by authorized users with password. Three different password levels are available to authorize selective access: user level, advanced level, expert level. The whole advanced functions authorized to users with expert level password are described below.

3.4.2.1. Password management

The « expert » user only can activate this function by touching/double-clicking on  to display the password management pop-up.



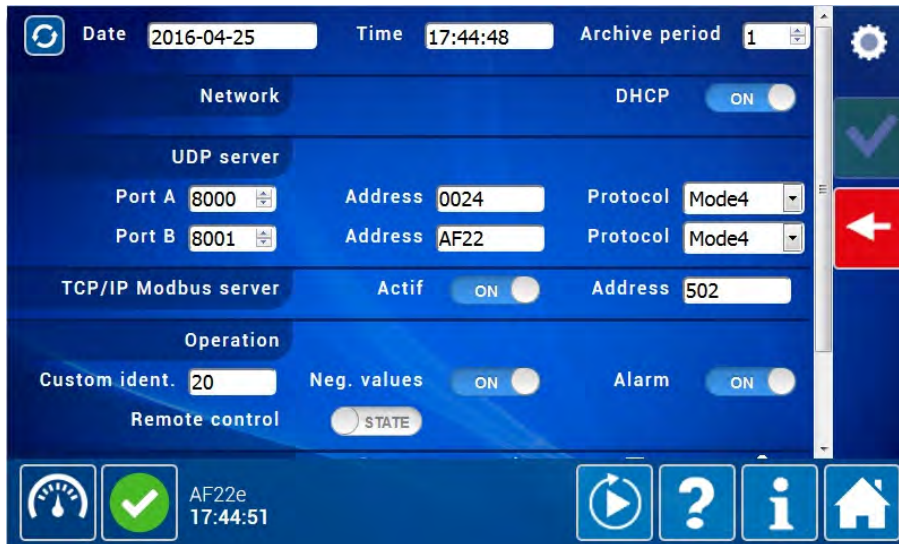
To modify passwords, the user fill-in the corresponding fields, then touch/click-on  to validate modifications and close the window.

Touch/click-on  to close the window without validating.

3.4.2.2. General configuration

This screen is accessed by touching/clicking-on .

This screen is used to configure the addresses and protocols necessary for communication and the measurement channels.



DATE and TIME fields: they indicate the current date and hour of the analyzer. They are modifiable with the « expert » level password only.

 : This icon is displayed when the analyzer is connected to a PC or a tablet. By clicking on it, the user updates the date and hour of the analyzer with the current date and hour of the PC or the tablet.

ARCHIVE PERIOD field: it indicates the analyzer archive period. It is modifiable with the « expert » level password only.

DHCP field: it is used to activate the automatic network configuration.

When ON, it allows to connect the analyzer to a TCP/IP network whose address is generated by a DHCP server.

When OFF, the ADDRESS, MASK and GATEWAY fields are displayed to allow the user to define the TCP / IP address, gateway and mask necessary for connection. This can be done the « expert » level password only.

WIFI/WPA field: it indicates the WPA encryption key of WiFi. It is modifiable with the « expert » level password only (8 to 63 characters, 0-9 and A-F authorized).

UDP SERVER section: it allows configuration of the two available UDP ports, Port A and Port B.

- Port number can be fixed within the range 1000 to 9999.
- Address is used only for Mode4 (four alphanumeric characters) and JBUS (number 0000 to 0255).

They are modifiable with the « expert » level password only

OPERATION section:

- NEG. VALUES field ON/OFF indicates if negative values are authorized or not.
- ALARM field ON/OFF enables or disables the alarm management.

MEASUREMENT CHANNEL section: it is used to configure the measurement channel display.



3.4.2.3. Automatic cycle configuration

This screen is accessed by touching/clicking-on .

This screen allows configuration of the four periodic cycles of the analyzer: Zero, Span, Zero-Ref., and Calibration. These cycles can be triggered in automatic or manual mode.



INLET field: allows selecting the channel used for the cycle (Zero, Span, Ref-Zero, Calibration).

PROGRAMMED INLET field: when ON, the analyzer uses the programmed inlet when a cycle is launched manually.

REMOTE CONTROL field: allows triggering cycle on a remote-control input (when optional ESTEL board(s) is/are available...).

CYCLIC field: enables or disables cycle triggering in automatic mode.

DURATION field: allows fixing cycle time.

PURGE DURATION field: it fixes purge time. Purge is applied before or after the cycle depending on the analyzer.

PERIOD field: allows fixing periodicity of automatic triggering.

START HOUR field: allows fixing start hour to trigger cycles automatically. If several cycles are in automatic mode, they will be triggered in the following order: Zero-Ref, Calibration, Zero, and Span.

AT STARTUP field: allows to force a Zero-Ref cycle triggering at analyzer startup (when warming-up cycle is finished).

3.4.2.4. Language selection

The available languages are English and French.

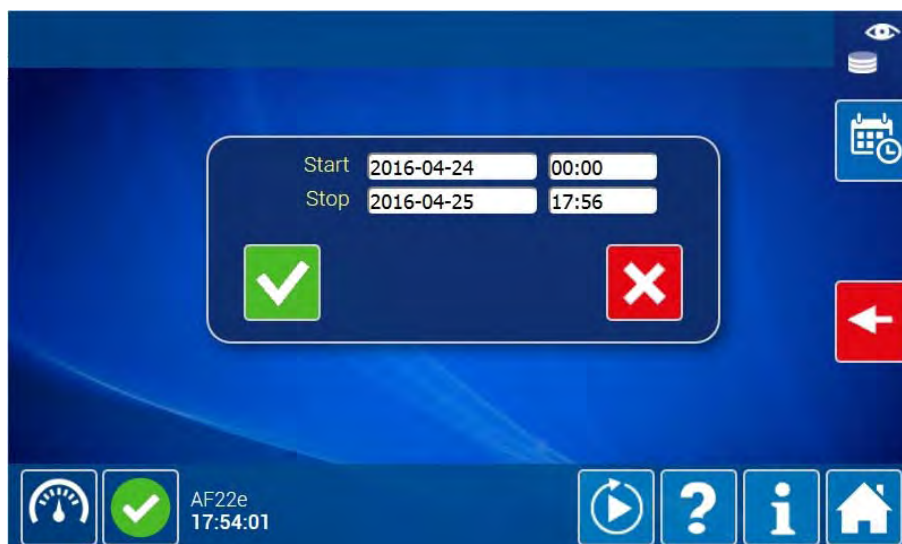
The user touches/double-clicks on  to display the language selection pop-up. The flag surrounded by a red frame indicates the language in use. The user touches / clicks-on the other flag to select the desired language and close the pop-up.

The message « Language selection » is displayed at screen top when English is selected, and « Choix de la langue » when French is selected.



3.4.2.5. Find and display recorded averages

This screen is accessed by touching/clicking-on . It allows to set the data period the user wants to watch, and to display these data.



The user touches/clicks-on  to validate the selection, close the pop-up and display the screen to visualize the recorded averages for the period. Touch/click-on  to close the pop-up without validating modifications.



Button definition specific to this screen



: exports the displayed data to a TXT file.



: opens the selection popup of dates and times period the user wants to view.

Icon meaning



: indicates an average stored with an alarm status. Alarm details are displayed by touching this icon.



: indicates an average stored with a warning status. Warning details are displayed by touching this icon.



: indicates an average stored with a "calibration" status. Calibration details are displayed by touching this icon (zero, span).

3.4.2.6. Delete all recorded averages

Touching/ double clicking-on this button  delete all the recorded averages.

WARNING : **this action is irreversible.**

3.4.2.7. List of connected clients

The user touches/double-clicks on  to display the pop-up giving the IP addresses and inactivity duration of the currently connected clients.

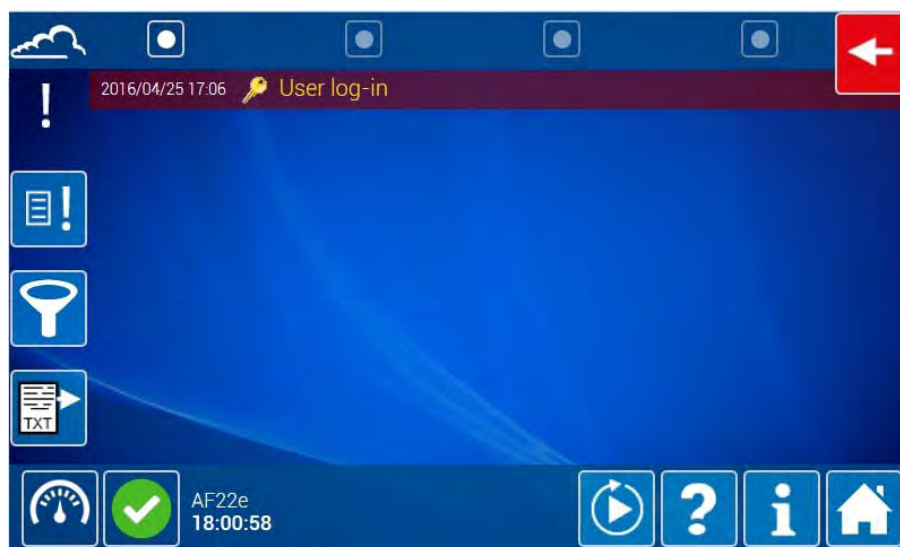


Click-on/touch  to close the pop-up.

3.4.2.8. Diagnostic functions

This screen is accessed by touching/clicking-on . It consists of four pages displayed by touching/clicking-on the white points in the screen top browsing bar.

The first page lists the current alarms and significant events on the analyzer and allows access to page 4 for alarm solving.



Button definitions specific to this screen



Displays or hides all the archived events.



Displays or hides the simple events. It acts as filter.



Exports the history of the recorded events in a TXT file.

The second page displays the analyzer signal values:

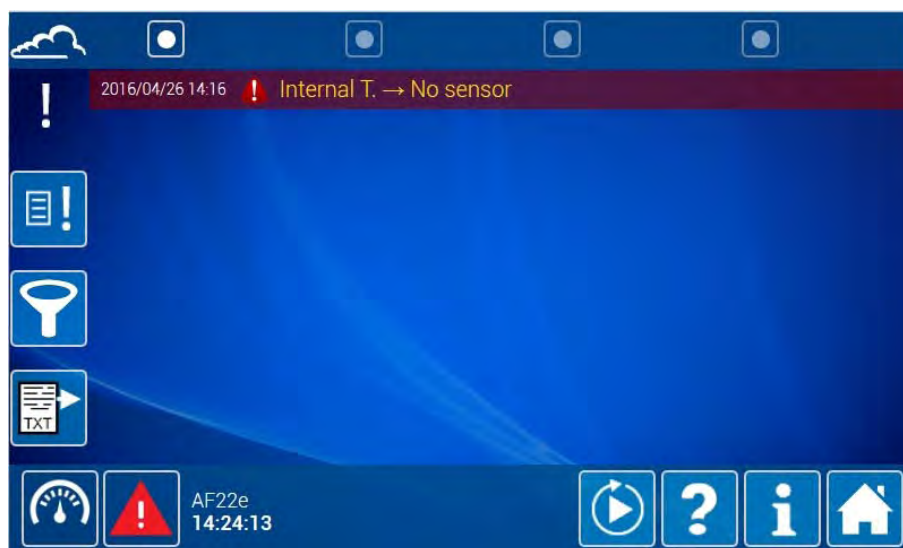


The third page displays the control status, the component list and status when detected on analyzer startup. The ON/OFF buttons are used to check that the corresponding elements are working correctly. When the user exits the screen, the elements returns to their standard status.



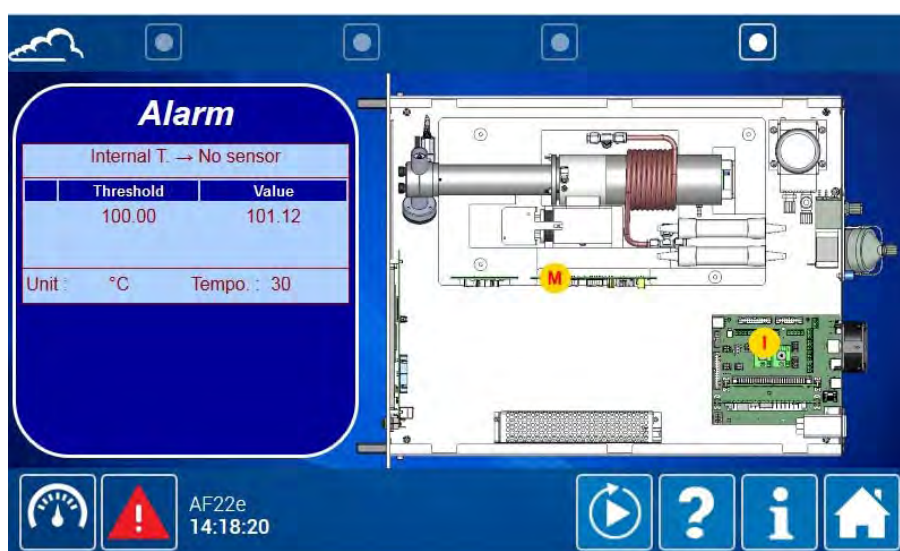
Alarm solving :

When an alarm triggers, the alarm icon  is displayed at the bottom left side of the screen. The user touches/clicks-on to access the diagnostic function first page displaying the current alarm and event list, as shown below: fields indicate triggering hour and fault nature. The user touches/clicks-on  to access the diagnostic function fourth page.

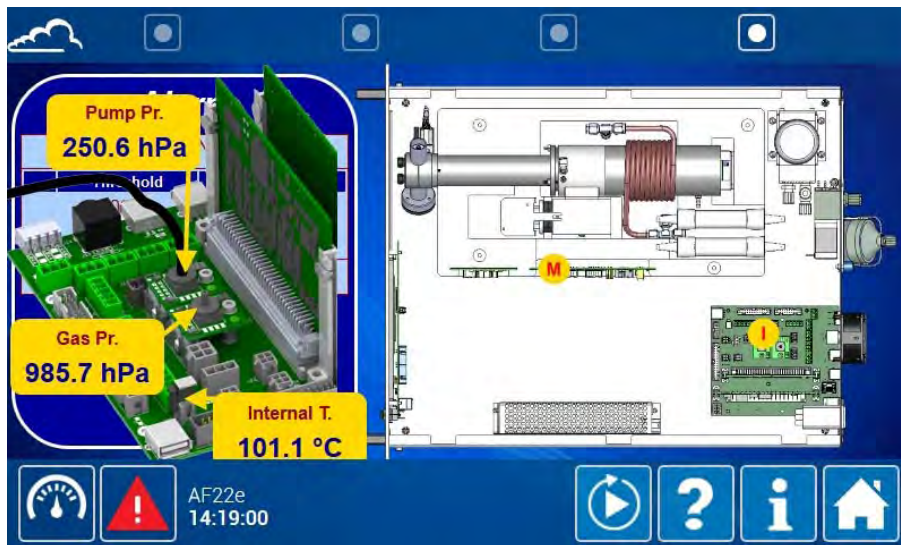


The fourth diagnostic function page displays the alarm threshold exceedance values, and shows the concerned faulty element in the internal analyzer view.

As example, the view below indicates a gas temperature alarm due to a sensor fault on the red-outlined board and focused with .



The user touches/clicks-on  to display the screen explaining what to do for alarm solving : the pressure and temperature values are written in the frame of internal view left side, as shown below :



When the alarm is cleared, the analyzer returns to its normal status, and the icon  replaces the icon



in the bottom left side.

3.4.2.9. Set the analyzer in standby mode

To activate stand-by mode, the user double-clicks on/touches  which becomes .

Standby mode is used to stop the pump while all the other controls remain in operation. To reactivate the measurement mode, double-click / touch .

When the analyzer is switched to standby mode, the status icon (in bottom left side of screen) is modified to  indicating standby status. However, alarm or control indication is a priority.

Consequently, if the analyzer is switched to standby mode while in alarm or control is active, the alarm icon  or control icon  will remain displayed and the standby icon  will not be displayed.

Under these conditions, to verify if the measurement or standby status is active, the user checks the button  is red-outlined or not in the advanced function page.

3.4.2.10. Set the analyzer in maintenance mode

To activate maintenance mode, the user double-clicks on/touches  which becomes .

Maintenance mode activation is used to flag that the analyzer is in maintenance. When maintenance mode is activated, the recorded data are tagged with a status mark: "Maintenance" appears next to the concerned data when they are downloaded.

Double-click on/touch  to reactivate the measurement mode.

As the analyzer is switched to maintenance mode, the status icon in the left bottom side of the screen changes to  indicating maintenance status. However, alarm or control indication is a priority. Consequently, if the analyzer is switched to maintenance mode while in alarm or control is active, the

alarm icon  or control icon  will remain displayed and the maintenance icon  will not be displayed.

Under these conditions, to verify if the measurement or maintenance status is active, the user checks the button  is red-outlined or not in the advanced function page.

3.4.2.11. Advanced analyzer configuration

This screen is accessed by touching/clicking-on . This screen allows configuration of the metrological parameters.



COMPOUND SO₂ section:

- « Unit » field: selects the compound measurement unit.
- « Threshold 1 » and « Threshold 2 » fields: fix a limit value to the measurement. If measurement value is out of this limit, an alarm triggers.
- « Conversion Coef. » field: displays the coefficient used for unit conversion (for ex. : ppb → µg/Nm³). This function is only applied if unit is µg/m³, mg/m³, g/m³.
- « Offset » and « Span Coef. » fields: these two values are used to calibrate the measurement with the linearization curve $Y = Ax + B$. A is the span coefficient, B is the offset. The span coefficient can be modified by the user or by a calibration cycle.
- « Zero adjust » field: allows setting manually the compound baseline (zero).
- « Delta (%) » field: when the span coefficient is modified by a calibration cycle, this value indicates the difference between the new coefficient and the previous coefficient. If this value is greater than 50%, a span alarm is triggered. It is necessary to re-validate manually the span coefficient to disable this alarm. When the analyzer is in alarm, it does not take account the new calculated calibration coefficient, but keeps the previous calibration coefficient.

« *Span gas value* » field: span gas cylinder concentration used to perform calibration check. This concentration is the reference value to be reached.

« *Calib. ON/OFF* » field: this field is used when the analyzer can measure many gases. It allows to perform a gas-by-gas calibration with no effect on the other gases. It invalidates or not the auto-calibration for the considered gas.

« *Response time* » field: shows response time value. Refer to Chapter 2 of this manual for more details.

« *Filter autonomy (d)* » field: shows the new zero air filter autonomy (day number). The analyzer decrements each day from this value (refer to synoptic display of the standard function screens). A control triggers when it lasts 120-day autonomy, and an alarm is triggers when it lasts one-day autonomy.

« *Kicker autonomy* » field (d): shows the new carbon kicker filter autonomy (day number). The analyzer decrements each day from this value (refer to synoptic display of the standard function screens). A control is triggers when it lasts 120-day autonomy, and an alarm is triggers when it lasts one-day autonomy.

« *Init. Counter OFF/ON* » field: reset filter autonomy to the programmed value after a filter change.

« *Dilution OFF/ON* » field: used to activate the dilution function and to set the dilution coefficient.

3.4.2.12. Automatic linearization of the flow

This screen is accessed by touching/clicking-on . It allows to linearize the flow value as function of the corresponding pump set point.

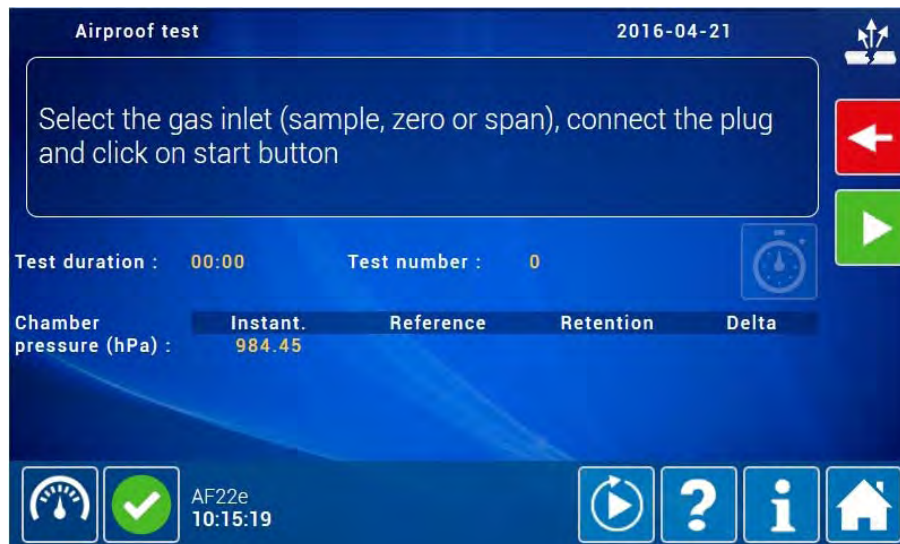
Before linearization launching, the user must connect a flowmeter (0-100 nl/h) to the suitable fluid inlet. Then, he clicks-on/touches the corresponding fluid inlet button , or , or , and he inputs in the « Nominal » field the read value on the flowmeter. Then, he launches linearization by touching/clicking on . Then, the analyzer will be in a maintenance mode for around 5 minutes.

REMARK: the in-factory pump setting is 100% by-default.



3.4.2.13. Airproof test

This screen is accessed by touching/clicking-on . It explains the procedure to follow for performing the airproof test of the analyzer fluid circuit. The user must follow the instructions detailed in the white-outlined frame.



When the user launches the airproof test, the analyzer switches to maintenance mode.

3.4.2.14. Potentiometer setting

This screen is accessed by touching/clicking-on .

It is used to check potentiometer setting of the PM preamplifier board.



3.5. CALIBRATION

IMPORTANT NOTICE:

Analyzer use, as an equivalent method for EPA reporting, requires periodic multipoint calibration and subsequent zero/span checks as described below. All gases for calibration must be traceable to a National Institute of Standards and Technology (NIST) reference.

3.5.1. OVERVIEW OF CALIBRATION AND CONCEPTS

To ensure the accuracy of performed measurements using the AF22e analyzer, the unit must be regularly checked, calibrated and adjusted, following the user's quality assurance plan.

- Zero and span check :

This operation compares the analyzer response, for the zero air and span point, to the used gas standards.

This check is used to measure the analyzer drift in time without modifying the « Span Coeff. ».

This check can be performed by using the internal zero air and the span gas.

Periodicity: generally 24 hours, in automatic cycle mode for the zero check.

- 2-point calibration:

This procedure is used for checking and correcting the analyzer response to zero and span points located at approximately 0 % and 80 % respective of the full scale of the measurement range used.

Periodicity: monthly, or more frequently if the installation requires it.

- Span (multi-point calibration):

This involves a complete checkup of the analyzer's linearity performance.

Periodicity: quarterly, or following out-of-tolerance calibration check results requiring analyzer intervention.

Note about gas cylinder connections:

When the analyzer is equipped with an internal zero filter and / or an internal permeation bench, calibration gas cylinders will be connected to the sample inlet.

Note about gas generation devices:

For pressurized gas devices, it is necessary to provide an excess system to deliver gas at atmospheric pressure to the analyzer inlet. The device materials should be neutral for the gas used. When the device is used in automatic cycle with a cylinder, it is necessary to provide a shutoff solenoid-valve remote-controlled by the analyzer (refer to Figure 3–3).

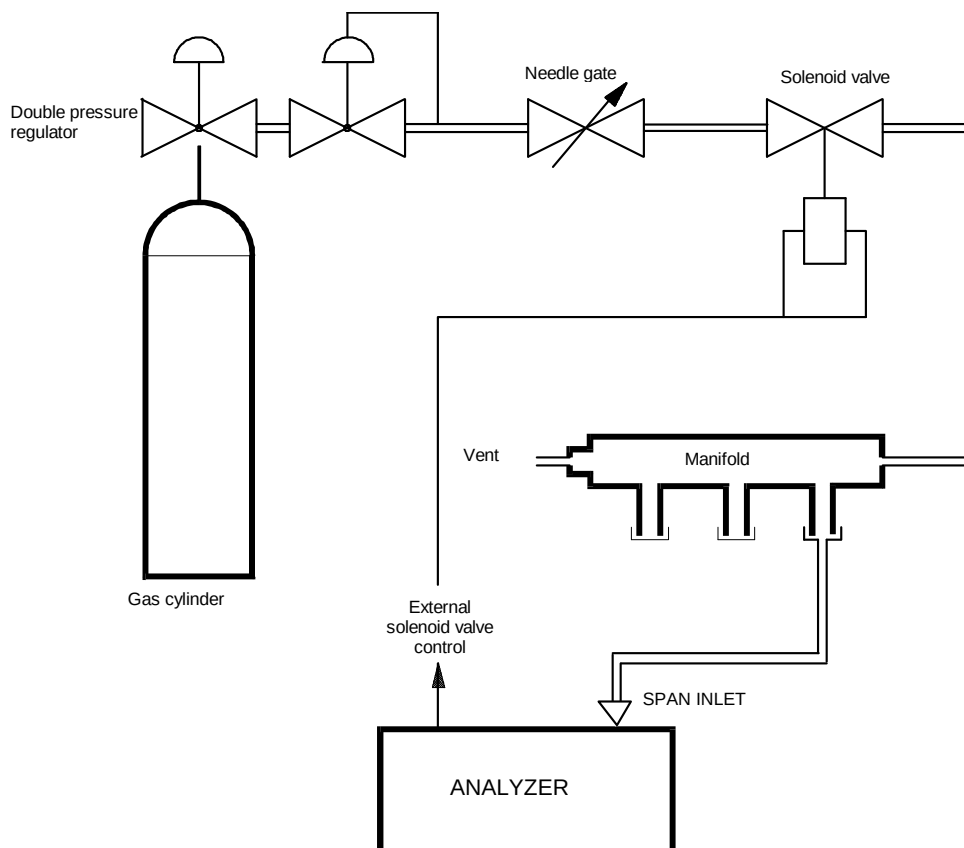


Figure 3–3 – Example of a pressurized gas connection

3.5.1.1. Zero air generation

For check and calibration, zero air generation can be obtained using the following methods:

- Zero check:

A sufficient quality of zero air is obtained using the analyzer zero air filter (Purafil/activated vegetable charcoal).

- Calibration:

Cylinder of reconstituted air or zero air generator.

Zero air should not contain more than 0.5 ppb of SO₂, and be free of contaminants likely to be measured by the analyzer.

3.5.1.2. Span gas generation (SO₂)

NOTE : The span gas cylinders used should be certified following the user's quality assurance plan.

The zero air used for the standard gas dilution and for the analyzer zeroing should be the same. It should contain less than 0.5 ppb of SO₂.

When a permeation device is used, the appropriate warm-up duration is 24 hours.

The devices used for span gas generation will deliver 25 L/h minimum and 60 L/h maximum.

All fittings, valves, pneumatic lines and other elements in contact with the span gas have to be made of highly SO₂-inert materials (i.e. PTFE, glass or stainless steel).

- Scale point check :
 - SO₂ cylinder (≤ 2 % accuracy) with a concentration of approximately 80 % of the full scale of the measurement range used.
 - Dynamic generation of one scale point using a cylinder and a diluter to generate approximately 80 % of the full range scale.
 - Optional built-in permeation oven with SO₂ tube. The concentration generated by this device is recorded on the test report (for span check only).
 - External permeation device equipped with SO₂ permeation tube to generate approximately 80 % of the full range scale.
- 2-point calibration :
 - SO₂ cylinder (≤ 2 % accuracy) with a concentration of approximately 80 % of the full scale of the measurement range used.
 - Dynamic generation of one scale point using a cylinder and a diluter to generate approximately 80 % of the full range scale.
 - External permeation device equipped with SO₂ permeation tube to generate approximately 80 % of the full range scale.
- Multi-point calibration :
 - A cylinder of SO₂ span gas in nitrogen (N₂) or air, and a dilution in zero air must be provided to obtain the various required concentrations. It is advised to use a zero air dilution system providing a 100-time higher flow than SO₂-in-nitrogen cylinder flow. Many certified cylinders at the required concentrations can also be used.

NOTE : The span gas cylinders used should be certified following the user's quality assurance plan.

3.5.1.3. Internal solenoid valves

When performing a multipoint calibration, the user should connect the zero and span gas sources to the sample inlet port. After this calibration, the zero and span sources should be connected to the corresponding analyzer inlet ports. The AF22e should give identical responses whether the sources are connected to the sample or the zero and span inlets ports. If not, then internal valves should be serviced.

The various gas inlet ports can be used for zero and span checks and two-point calibrations.

3.5.2. ZERO AND SPAN POINT CHECK

3.5.2.1. Equipment Required

- Zero check:
Use method described in section 3.5.1.1.
- Span point:
Use method described in section 3.5.1.2.

3.5.2.2. Procedure

Pay attention: touching  allows to display ,  and . Touching  allows to display  and .

- Zero check:
Check that internal zero filter autonomy is higher than 30 days, or connect a zero air free of SO₂ such as a 5.0 N₂ cylinder (99.999% ultra-high purity N₂ cylinder, i.e. containing less than 10 ppm of total impurities).
Use the  key to select zero. Wait for the measurement to stabilize. The measurement should be within ± 2 ppb (taking account the offset if it is programmed). If not, a zero calibration is necessary.
- Span check:
Use the  key to select the span inlet port. Wait for measurement to stabilize. Result will be compared to the concentration generated by the device used, taking account its accuracy as well as a possible programmed offset.

3.5.2.3. Use of automatic cycles

To program the cycles, touch  to access the « *Automatic cycle configuration* » screen (refer to section 3.4.3.2).

- Zero cycle:
The zero air filter is permanently connected to the analyzer zero air inlet port. The recommended Zero check duration is 600 seconds, as indicated in the Duration field of the Cycles / Zero column in the screen shown below.
- Calibration cycle:
The span gas generation device is permanently connected to the analyzer span gas inlet port and the span inlet is selected for span check. The SO₂ concentration must be below the full scale of the measurement range used. The recommended check duration is 600 sec, as indicated in the Duration field of the Cycles / Span column in screen shown below.

Cycles	Zero	Span	Ref-Zero	Calibration	
Inlet	Zero	Span	Zero	Span	
Programmed inlet			ON	ON	✓
Remote control	ON	ON	OFF	OFF	←
Cyclic	OFF	OFF	OFF	OFF	
Duration	600 s	600 s	600 s	600 s	
Purge duration	0 s	0 s	0 s	10 s	
Period	6 h	24 h	24 h	24 h	
Start hour	15:00	00:00	00:00	00:00	
At startup			ON		

AF22e
03:19:24

3.5.3. TWO-POINT CALIBRATION

3.5.3.1. Equipment required

- Zero reference :
Use method described in section 3.5.1.1.
- Span point:
Use method described in section 0.

3.5.3.2. Procedure

- Zero calibration:
 - Use the  key to start a zero-reference cycle on the zero inlet port (minimum 600 sec). Wait for measurement to stabilize. Measurement should be within ± 0.5 ppb (taking account the offset if it is programmed). If this is not the case, it is necessary to change the internal zero filter load (refer to maintenance sheet in section 4).
- Span point correction:
 - Automatic correction:
Use the  key to start a calibration cycle. The analyzer switches automatically to the programmed inlet port if the optional programmed inlet is set to ON. Otherwise, use the  key to select span inlet or the  key to select sample inlet before starting a calibration cycle.
The analyzer automatically modifies its calibration coefficient as a function of the programmed gas concentration. Recommended duration for self-calibration is 600 seconds.
 - Manual correction:
Use the  key (span inlet) or the  key (sample inlet) to select the appropriate inlet to which the gas is connected. Wait for the measurement to stabilize. Set the desired K in « Calibration coef. » field of « *Advanced analyzer configuration* » screen, as follows:

$$K_{new} = K_{former} \times \frac{\text{Span gas value}}{\text{Read value}_{(\text{without offset})}}$$

For details about the calculation of span gas values and read values, refer to section 3.4.3.4.

CAUTION: It is important to note the former K reading before modifying it, since it will be deleted when the new K validation is done.

3.5.4. MULTI-POINT CALIBRATION

3.5.4.1. Overview

An appropriate device consists of a diluter (MGC101 type), a certified span cylinder (SO_2) (2 % accuracy), and a zero air generator. Certified span cylinder calibration may be associated to reference materials of the National Institute of Standards and Technology (NIST) for calibration in the US EPA measurement field. Refer to Figure 3–4.

Gases will be applied at atmospheric pressure to the analyzer sample gas inlet.

Analyzer calibration requires gas generation of 7 points including zero (example: 0, 15, 30, 45, 60, 75 and 90% of the full scale of the measurement range used). The dilution carrier gas must be the same as for zero measurement.

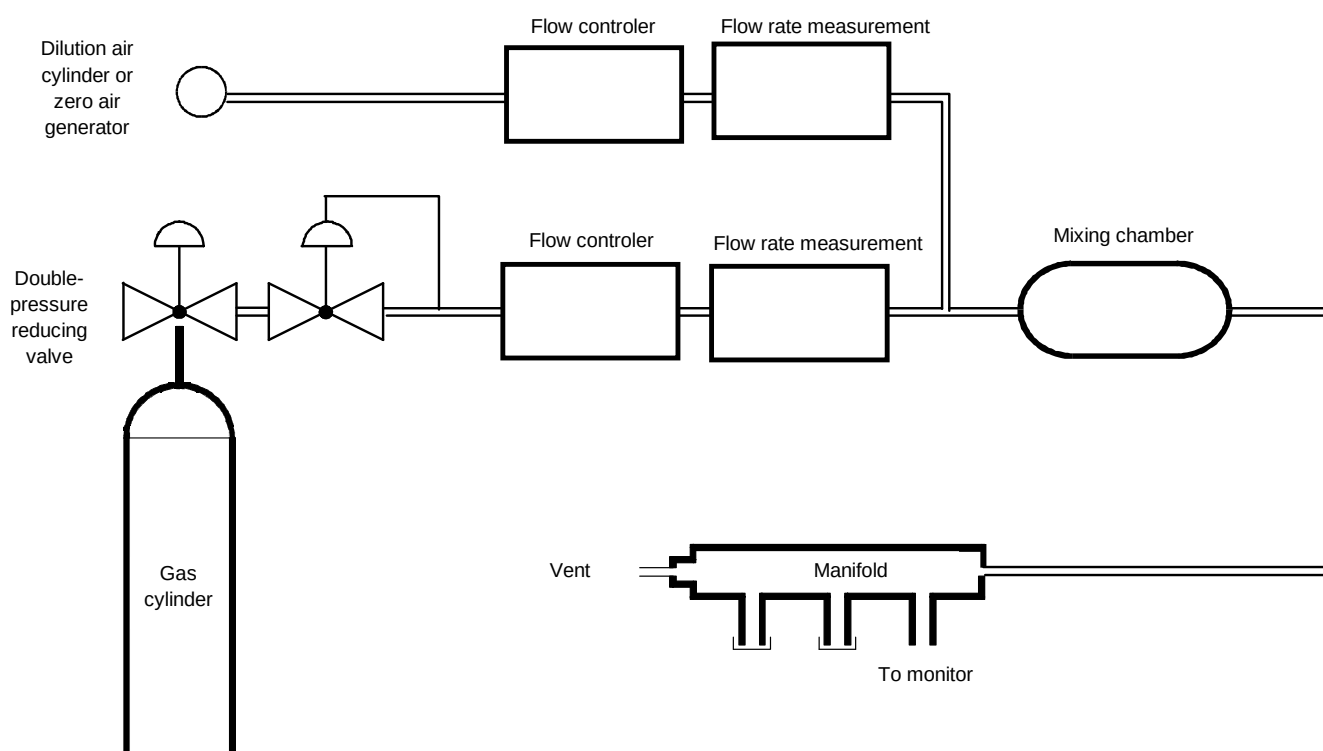


Figure 3–4 – Typical calibration device diagram

3.5.4.2. Equipment Required

- Diluter:
 - Flow controllers: should regulate flow rates to ± 1 %.
 - Flow meters: should allow flow measurement and recording to ± 2 %.
 - Mixing chamber: shape and size should allow homogeneous mixing of span gas and dilution air.
- Manifold:

The manifold includes at least one excess flow outlet and one “analyzer” outlet. This “analyzer” outlet has a sufficient diameter to avoid load losses at the analyzer inlet. The excess flow outlet (at least 20 % of the total flow rate) is designed so the manifold pressure will be close to atmospheric pressure (no overpressure) and ambient air should not cause back pressure.

- Dilution air (zero air):

Zero air generator or cylinder of reconstituted air.

Zero air should not contain more than 0.5 ppb of SO₂ and be free of contaminants that might be measured by the analyzer.

- Span gas (SO₂):

Certified SO₂ cylinder, concentration 2% in air allowing six-range generation between 15 % and 90 % of the full scale of the measurement range used. Carrier gas may be nitrogen if the system allows a higher dilution ratio than 600:1.

3.5.4.3. Procedure

- Turn on the analyzer at least 6 hours before performing calibration.

- Touch the  key to display « Automatic cycle configuration » screen and program analyzer as follows (advised configuration) :

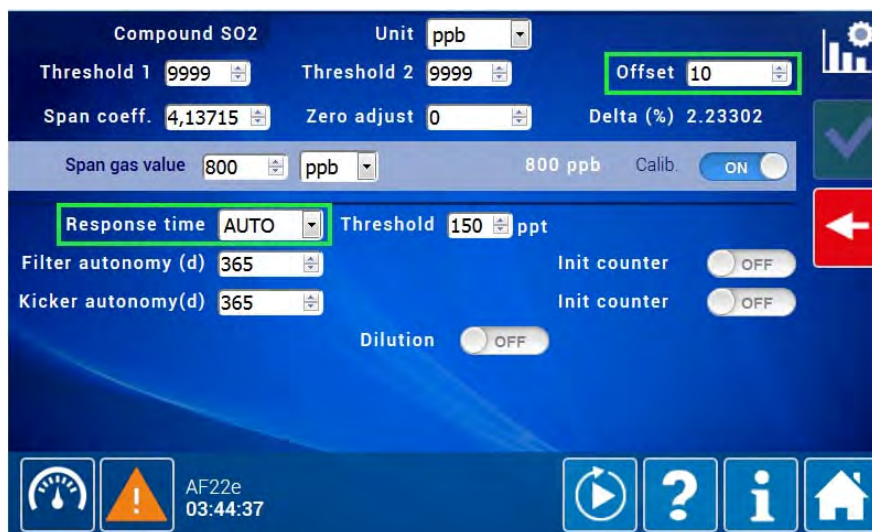


The screen displays configuration options for four cycles: Zero, Span, Ref-Zero, and Calibration. Each cycle has an 'Inlet' dropdown (Zero, Span, Zero, Span) and a 'Programmed inlet' toggle (OFF). Below these are 'Remote control' and 'Cyclic' toggles (OFF). 'Duration' is set to 600 s for all. 'Purge duration' is 0 s for Zero and Span, and 10 s for Ref-Zero and Calibration. 'Period' is 24 h for all. 'Start hour' is 00:00 for all. 'At startup' is set to ON. The bottom status bar shows 'AF22e' and '03:39:03'.

- Touch the  key to display « Advanced analyzer configuration » screen:

Response time: set Auto

- Configure SO₂ offset to 10 ppb at least, or 5 % of the full scale of the measurement range used.



The screen displays configuration for Compound SO₂. 'Unit' is ppb. 'Threshold 1' and 'Threshold 2' are 9999. 'Offset' is 10 (highlighted with a green box). 'Span coeff.' is 4,13715. 'Zero adjust' is 0. 'Delta (%)' is 2.23302. 'Span gas value' is 800 ppb. 'Calib.' is ON. 'Response time' is AUTO (highlighted with a green box). 'Threshold' is 150 ppt. 'Filter autonomy (d)' and 'Kicker autonomy (d)' are 365. 'Init counter' is OFF. 'Dilution' is OFF. The bottom status bar shows 'AF22e' and '03:44:37'.

- Connect analyzer sample inlet to manifold dilution system.
 - Adjust the dilution system in order to generate a total zero air flow higher than the analyzer sample flow rate by 20%.
 - Generate zero air. Wait for measurement to stabilize (recommended duration: 600 sec). Note Z_{SO_2} value, as a percentage of selected full scale, named ECH.
 - Follow steps in section 3.5.3.2. (« span point correction » section) to adjust calibration coefficient.
- ECH is used to name the analyzer full scale.

- SO_2 measurement : use formula below to calculate generated span value:

$$[SO_2]_{generated} = \frac{[SO_2]_{cylinder} \times F_{SO_2}}{F_D + F_{SO_2}} \quad (1)$$

Where:

$[SO_2]_{Generated}$ is SO_2 concentration of generated gas at manifold outlet,

$[SO_2]_{cylinder}$ is SO_2 concentration of certified cylinder,

$[F]_{SO_2}$ is SO_2 flow in NI/min,

$[F]_D$ is dilution air flow in NI/min.

To calculate read value (in ppb) to be taken into account, use following formula:

$$[SO_2]_{read} = \left(\frac{S_{record} - Z_{SO_2}}{100} \right) \times ECH \quad (2)$$

Where:

S_{record} is the recorder value, in percentage of recorder full scale,

ECH is the analyzer full scale,

Z_{SO_2} was measured previously.

- Then generate five SO_2 concentrations within 15 and 90 % of the full scale. For that, vary $[F]_{SO_2}$ and/or $[F]_D$ flows.
- Plot $[SO_2]$ read values as a function of $[SO_2]_{generated}$ values, including zero air point. Check linearity.
- Plot, or calculate using the least squares formula, the calibration line of the analyzer.

$$[SO_2]_{read} = a \times [SO_2]_{generated} + b \quad (3)$$

Where:

a : is the linear regression coefficient (slope), and is calculated as follows:

$$a = \frac{n \times \sum [SO_2]_{generated} \times [SO_2]_{read} - \sum [SO_2]_{generated} \times \sum [SO_2]_{read}}{n \times \sum [SO_2]_{generated}^2 - (\sum [SO_2]_{generated})^2} \quad (4)$$

b : is the constant linear regression term (intercept), it is calculated as follows:

$$b = \frac{\sum [SO_2]_{read} - a \times \sum [SO_2]_{generated}}{n} \quad (5)$$

n : is the data number.

CHAPTER 4

PREVENTIVE MAINTENANCE

4.1	SAFETY INSTRUCTIONS	4-3
4.2	MAINTENANCE CALENDAR	4-4
4.3	MAINTENANCE OPERATION SHEETS	4-4
4.4	KITS AND SPARE PARTS SET FOR AF22e MAINTENANCE	4-17

Figure 4-1 – Sample inlet dust filter replacement	4-5
Figure 4-2 – Internal zero filter replacement	4-8
Figure 4-3 – KNF pump head dismantling	4-10
Figure 4-4 – Diaphragm replacement	4-11
Figure 4-5 – Valves and O-rings replacement	4-12
Figure 4-6 – Pump reassembly	4-13
Figure 4-7 – UV source assembly dismantling	4-14
Figure 4-8 – UV lamp change	4-15
Figure 4-9 – UV source reassembly	4-16

Page intentionally left blank

4 PREVENTIVE MAINTENANCE

4.1 SAFETY INSTRUCTIONS

The user must follow these safety instructions at all times:

- Always turn off the power supply when performing analyzer maintenance.
- Personnel should be trained in the proper operation of this equipment before attempting to operate it.
- The manufacturer shall not be responsible for any adverse outcomes resulting from the following:
 - Use of the monitor by unqualified service personnel.
 - Use of the monitor under conditions other than those specified in this document.
 - Use of components or accessories not manufactured by ENVEA, failure to use recommended parts may reduce the safety features.
 - Use of this equipment in a manner not approved by ENVEA, as it can cause harm to the equipment or operating personnel.
 - Inappropriate maintenance of the analyzer.
- A periodic inspection is required.

4.2 MAINTENANCE CALENDAR

By design, the AF22e requires very limited maintenance. However, the unit must be regularly serviced to ensure proper performance over time. The routine maintenance schedule shown below is an example, and this schedule can vary according to operating conditions.

Nature of operations	Frequency	Sheet N°
Sample inlet dust filter replacement	2 to 4 weeks	4.3.1
Optical, electric and fluid parameter check	1 month	4.3.2
Internal zero filter replacement	6 months	4.3.3
KNF pump maintenance	1 year	4.3.4
UV lamp replacement	2 years	4.3.5

Annual check

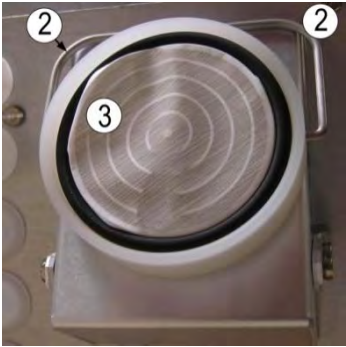
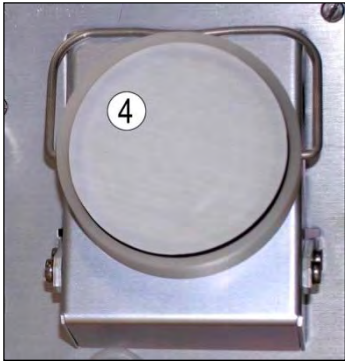
The monitor must undergo a thorough cleaning (measurement chamber, restrictors, flow circuit, etc.) in the laboratory at least once a year. All metrological parameters and fitting tightness should be checked at this time.

4.3 MAINTENANCE OPERATION SHEETS

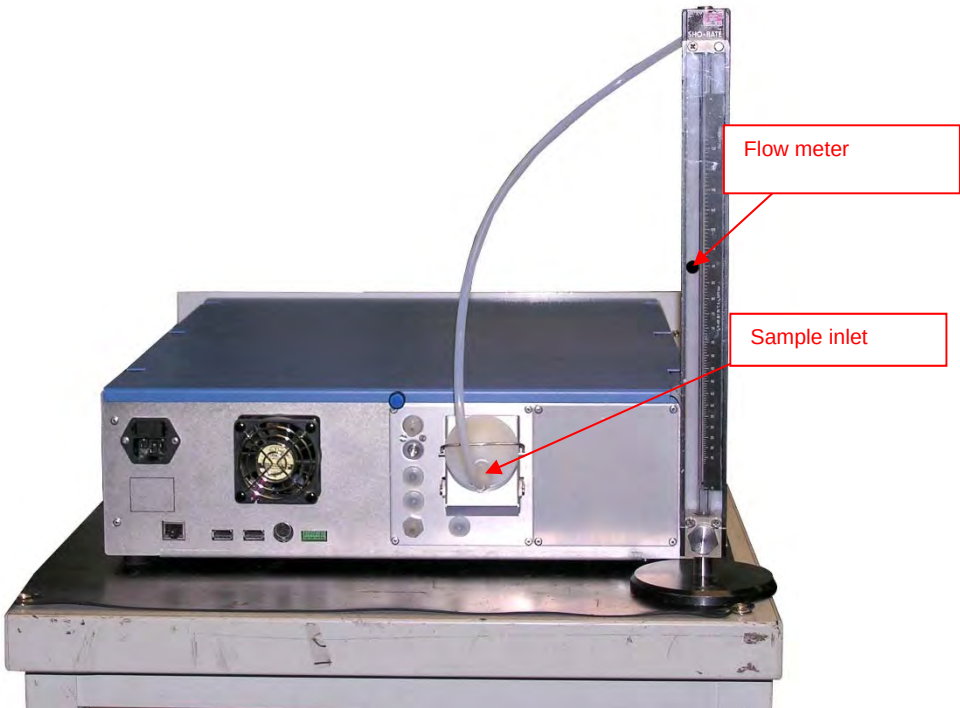
WARNING

- Do not use trichloroethylene or acetone to clean seals and gaskets.

MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.1	
Scope: Sample inlet dust filter replacement		PAGE: 1/1	Frequency: 2 to 4 weeks according to dust amount
<u>Proceed in the following order:</u> – Turn off the analyzer power switch and disconnect the sample inlet. – Unlock the filter holder cover (1) by pulling the spring (2) upwards. – Remove the worn filter (3). – Place the new filter (4) on the filter holder without the separating blue sheet. – Make sure that the filter correctly covers the filter-holder surface and that it is correctly placed flat on top of it. – Reposition the filter holder cover, and lock it by lowering the spring (2). – Connect the sample inlet and power on the analyzer.			Date
   (1) filter holder cover, (2) spring, (3) worn filter, (4) new filter. Figure 4-1 – Sample inlet dust filter replacement			
<u>Tools and parts required:</u> • Tweezers • Sample inlet filter (Teflon filter - porosity 5 µm - Ø 47 mm) - Ref.: F05-11-842			

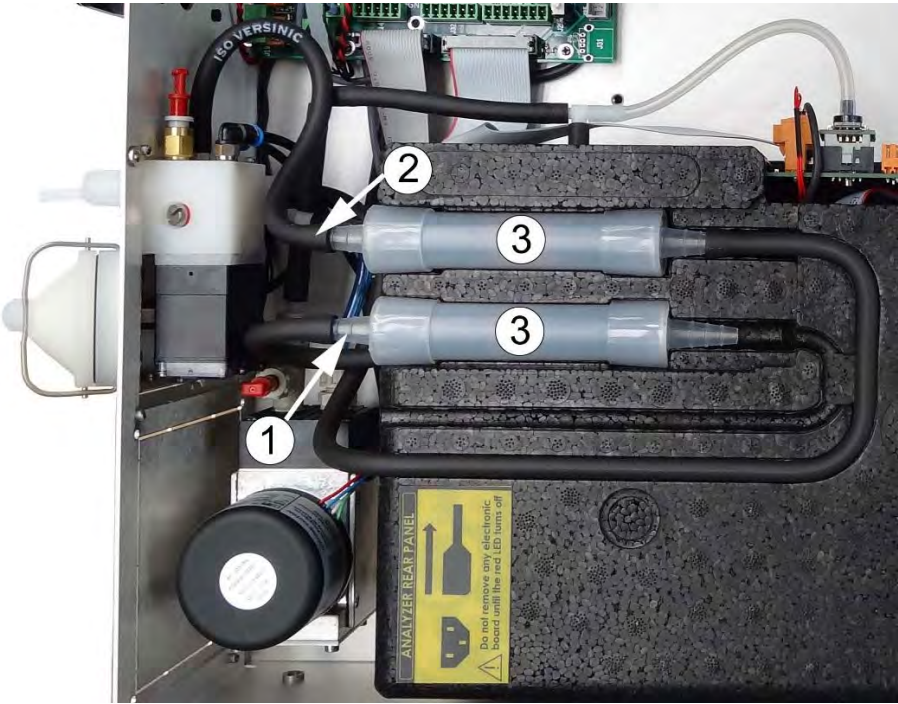
MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.2	
Scope: Optical, electric and fluid parameter check		PAGE: 1/2	Periodicity : 1 month
<u>Flow rate check :</u> Connect a flow meter on the AF22e sample inlet.			Date
			
Make sure the flow rate value indicated on the flow meter is 20 ± 5 NI/h.			
<u>Tools required</u> Flow meter, 100 liters/hour			

MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET : 4.3.2																																																									
Scope : Optical, electric and fluid parameter check		PAGE : 2/2	Periodicity : 1 month																																																								
<u>MUX signals check:</u> - Switch the analyzer to zero air. - Compare the values given on the « <i>Diagnostic functions (alarm, input/output, mux...)</i> » screen with the limits listed in the table below. <table border="1"> <thead> <tr> <th>Signals</th> <th>Min. values</th> <th>Typical values</th> <th>Max values</th> </tr> </thead> <tbody> <tr> <td>UV ref.</td> <td>250 mV</td> <td>N. A.</td> <td>N. A.</td> </tr> <tr> <td>HV PM</td> <td>1600 mV</td> <td>2800 mV</td> <td>4000 mV</td> </tr> <tr> <td>Internal T.</td> <td>5 °C</td> <td>30 °C</td> <td>50 °C</td> </tr> <tr> <td>Measurement chamber T°C</td> <td>48°C</td> <td>50 °C</td> <td>52 °C</td> </tr> <tr> <td>Gas pressure</td> <td>600 hPa</td> <td>990 hPa</td> <td>1200 hPa</td> </tr> <tr> <td>Pump pressure</td> <td>200 hPa</td> <td>300 hPa</td> <td>600 hPa</td> </tr> <tr> <td>+4V supply</td> <td>3950 mV</td> <td>4000 mV</td> <td>4050 mV</td> </tr> <tr> <td>+3.3 V supply</td> <td>3.10 V</td> <td>3.30 V</td> <td>3.50 V</td> </tr> <tr> <td>+12V supply</td> <td>11.50 V</td> <td>12 V</td> <td>12.60 V</td> </tr> <tr> <td>+24 V supply</td> <td>23.0 V</td> <td>24.0 V</td> <td>25.0 V</td> </tr> <tr> <td>+5 V supply</td> <td>4.70 V</td> <td>5 V</td> <td>5.30 V</td> </tr> <tr> <td>Sensor +5 V supply</td> <td>4.80 V</td> <td>5 V</td> <td>5.20 V</td> </tr> <tr> <td>+24V current</td> <td>- 0.1 A</td> <td>0.5 A to 3 A</td> <td>> 6 A</td> </tr> </tbody> </table>			Signals	Min. values	Typical values	Max values	UV ref.	250 mV	N. A.	N. A.	HV PM	1600 mV	2800 mV	4000 mV	Internal T.	5 °C	30 °C	50 °C	Measurement chamber T°C	48°C	50 °C	52 °C	Gas pressure	600 hPa	990 hPa	1200 hPa	Pump pressure	200 hPa	300 hPa	600 hPa	+4V supply	3950 mV	4000 mV	4050 mV	+3.3 V supply	3.10 V	3.30 V	3.50 V	+12V supply	11.50 V	12 V	12.60 V	+24 V supply	23.0 V	24.0 V	25.0 V	+5 V supply	4.70 V	5 V	5.30 V	Sensor +5 V supply	4.80 V	5 V	5.20 V	+24V current	- 0.1 A	0.5 A to 3 A	> 6 A	Date
Signals	Min. values	Typical values	Max values																																																								
UV ref.	250 mV	N. A.	N. A.																																																								
HV PM	1600 mV	2800 mV	4000 mV																																																								
Internal T.	5 °C	30 °C	50 °C																																																								
Measurement chamber T°C	48°C	50 °C	52 °C																																																								
Gas pressure	600 hPa	990 hPa	1200 hPa																																																								
Pump pressure	200 hPa	300 hPa	600 hPa																																																								
+4V supply	3950 mV	4000 mV	4050 mV																																																								
+3.3 V supply	3.10 V	3.30 V	3.50 V																																																								
+12V supply	11.50 V	12 V	12.60 V																																																								
+24 V supply	23.0 V	24.0 V	25.0 V																																																								
+5 V supply	4.70 V	5 V	5.30 V																																																								
Sensor +5 V supply	4.80 V	5 V	5.20 V																																																								
+24V current	- 0.1 A	0.5 A to 3 A	> 6 A																																																								
<u>Tools required</u> Flow meter, 100 liters/hour																																																											

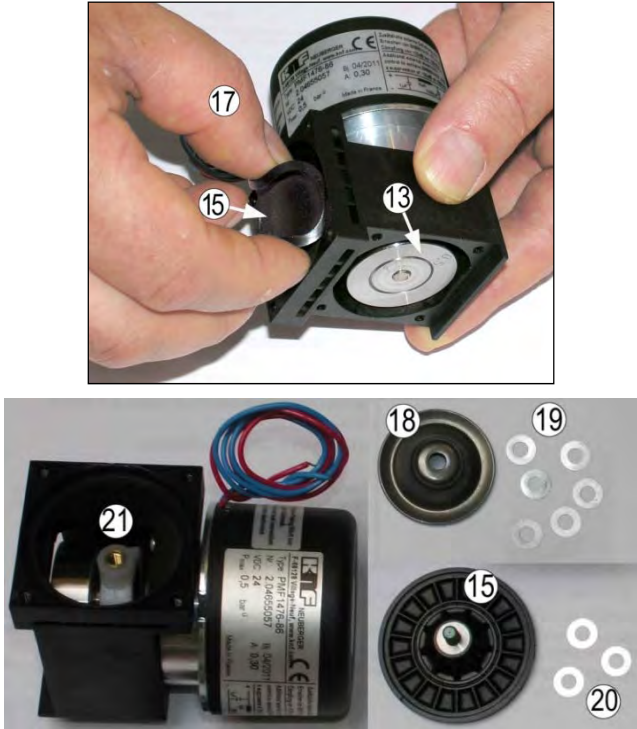
MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.3	
Scope: Internal zero filter replacement		PAGE: 1/1	Frequency: 6 months
Proceed in the following order: – Turn off the analyzer and unplug the power cable. – Remove the cover screw located on the analyzer rear panel. Remove the protective cover. – Disconnect the hydrocarbon scrubber tube (1) and the solenoid valve tube (2). – Remove the internal zero filters (3) and replace with new ones. – Reconnect the hydrocarbon scrubber tube (1) and the solenoid valve tube (2). – Plug in the power cable and power on the analyzer. – Click-on the alarm icon displayed in the bottom bar of any screen to view active alarms. Then, click-on the internal zero filter alarm and set the internal zero filter countdown to 180 days. – Wait at least one hour for the measurement to stabilize, then launch a new zero-reference cycle.			Dates
 (1) hydrocarbon scrubber tube, (2) solenoid valve tube, (3) internal zero filters. Figure 4-2 – Internal zero filter replacement			
Tools and parts required: • Internal zero air filter. Ref.: F05-0128-C-SAV			

MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.4	
Scope: KNF pump maintenance		PAGE: 1/5	Frequency: 1 year
WARNING: Before any intervention, unplug the power cable and make sure the pump is turned off. The diaphragm and valves are the only pump parts that need replacement. Maintenance operation consists of cleaning and/or replacing them. <u>Preliminary operation:</u> – Turn off the analyzer and unplug the power cable. – Unscrew the cover screw located on the analyzer rear panel. Remove the protective cover. – Disconnect the pump from its electrical connection. Unscrew the fluid inlet and the fluid outlet of the pump. – Unscrew and remove the four screws to release the pump holder out of the frame. – Unscrew and remove the three screws to release the pump from its holder, and perform pump maintenance outside the analyzer.			Date
<u>Tools and parts required</u> • Two flat screwdrivers, 1 x 5.5 cm and 0.4 x 2.5 cm • Hexagon socket head wrench n°3 • KNF pump maintenance kit. Ref.: V04-0005 • Cross-tip screwdriver, 4.5 x 75 cm			

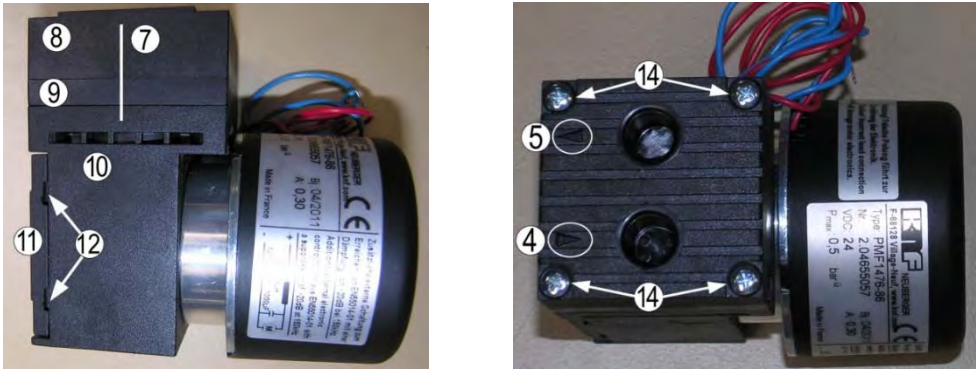
MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.4	
Scope: KNF pump maintenance		PAGE: 3/5	Frequency: 1 year
Diaphragm replacement: – Turn the eccentric wheel (13) to position the diaphragm (15) to the upper position, so the diaphragm can be easily grasped by its circumference (17), to unscrew turn counter clockwise. – The diaphragm is placed on a cupel (18). Remove the washers (19) located between the diaphragm and the cupel, and the washers (20) located between the cupel and the threading hole (21). Note carefully the washer number at each location because it is critical to install the same washer number at the same location during reassembly. – To change the diaphragm (15), put the washers (19) between the diaphragm and the cupel (18) back in place, and install the diaphragm on the cupel. Replace the washers (20) on the external cupel side so they are correctly in place between the cupel and the threading hole (21) upon reassembly. – Completely rescrew the assembly (diaphragm + cupel + washers) in the threading hole (21).			Date
 (13) eccentric, (15) diaphragm, (17) the user grasps the diaphragm by its circumference, (18) cupel, (19) washers, (20) washers, (21) threading hole. Figure 4-4 – Diaphragm replacement			
Tools and parts required • Two flat screwdrivers, 1 x 5.5 cm and 0.4 x 2.5 cm • Hexagon socket head wrench, n°3 • KNF pump maintenance kit. Ref.: V04-0005 • Cross-tip screwdriver, 4.5 x 75 cm			

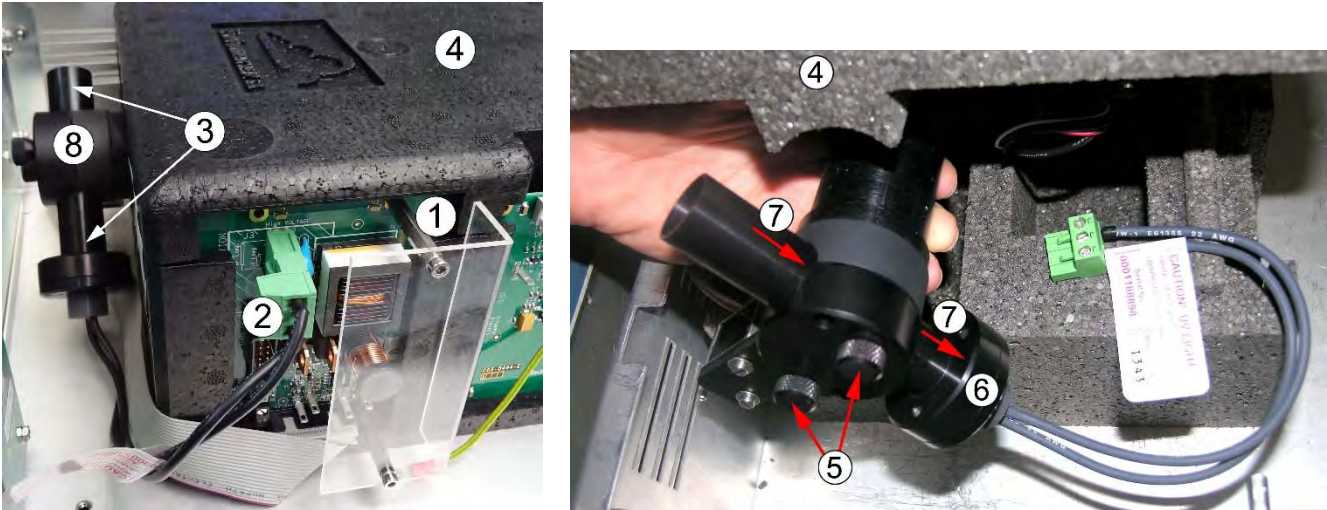
MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.4	
Scope: KNF pump maintenance		PAGE: 4/5	Frequency: 1 year
<u>Valves (22) and O-rings (23) replacement:</u> The valves are located in the cylinder head. Replace the valves (22) and the O-rings (23), taking care to correctly fit together the O-rings in their housings .			Date
(22) valves, (23) O-rings. Figure 4-5 – Valves and O-rings replacement			
<u>Tools and parts required</u> • Two flat screwdrivers, 1 x 5.5 cm and 0.4 x 2.5 cm • Hexagon socket head wrench, n°3 • KNF pump maintenance kit. Ref.: V04-0005 • Cross-tip screwdriver, 4.5 x 75 cm			

MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.4	
Scope: KNF pump maintenance		PAGE: 5/5	Frequency: 1 year
Pump reassembly: Reassembly is carried out in the reverse order from dismantling. - Reassemble the cover (8) and the cylinder head (9) on the pump body (10) according to the line drawn to mark the various parts (7). Put the four screws of the pump head (14) in place and then screw them clockwise. - Replace the eccentric cover (11). - Reassemble the pump on its holder, replace the screws, and tighten the screws. - Reassemble the pump with its holder inside the analyzer, replace the screws and tighten the screws. Reconnect the electric connector, the fluid inlet line, and the fluid outlet line. - Replace the analyzer cover, replace the screw, and power on the analyzer.			Date
 (4) fluid inlet, (5) fluid outlet, (7) pen mark, (8) cover, (9) cylinder head, (10) body pump, (11) eccentric cover, (12) slots, (14) pump head screws. Figure 4-6 – Pump reassembly			
Tools and parts required - Two flat screwdrivers, 1 x 5.5 cm and 0.4 x 2.5 cm - Hexagon socket head wrench, n°3 - KNF pump maintenance kit. Ref.: V04-0005 - Cross-tip screwdriver, 4.5 x 75 cm			

MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.5	
Scope: UV lamp replacement		PAGE: 1/3	Frequency: 2 years
WARNING: DO NOT put your fingers on the lamp glass. <u>Dismantling the UV source assembly:</u> – Turn off the analyzer and unplug the power cable. – Remove the cover screw located on the analyzer rear panel. Remove the protective cover. – Unplug the UV source supply board (3) connector (2) to disconnect the UV source assembly (1). – Slightly pull up the measurement chamber + UV source assembly to remove from the foam block (4), and loosen the two UV source holder screws (5). Then, handle the lower part (6) of the UV source assembly to pull it downwards as arrows (7) indicate. – Remove the UV source assembly (3) from its holder (8).			
 (1) UV source supply board, (2) UV source connector, (3) UV source assembly, (4) foam block, (5) UV source holder screws, (6) lower part of UV source assembly, (7) arrows, (8) UV source holder. Figure 4-7 – UV source assembly dismantling			
<u>Tools and parts required</u> • Screwdriver, 5.0 x 75 mm • UV lamp, Ref. : D01-1297			

MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.5	
Scope: UV lamp replacement		PAGE: 2/3	Frequency: 2 years

WARNING: DO NOT put your fingers on the lamp glass.

UV lamp replacement:

- Unscrew the three flange (9) screws (10) and pull off, as arrow indicates, to remove the UV lamp (11) from its protection tube (12).
- Replace UV lamp (11) with a new one.
- Insert the new UV lamp inside its protection tube (12) and take care to center the UV lamp window MAX OUTPUT side (13) with the protection tube window (14).
- Reassemble the flange (9) and tighten the three screws (10) to avoid UV lamp moving in its protection tube (12).

(5) UV source holder screws, (8) UV source holder, (9) flange, (10) three screws, (11) UV lamp, (12) UV lamp protection tube, (13) UV lamp window, (14) protection tube window.

Figure 4-8 – UV lamp change

Tools and parts required

- Screwdriver, 5.0 x 75 mm
- UV lamp, Ref. : D01-1297

MAINTENANCE SHEET

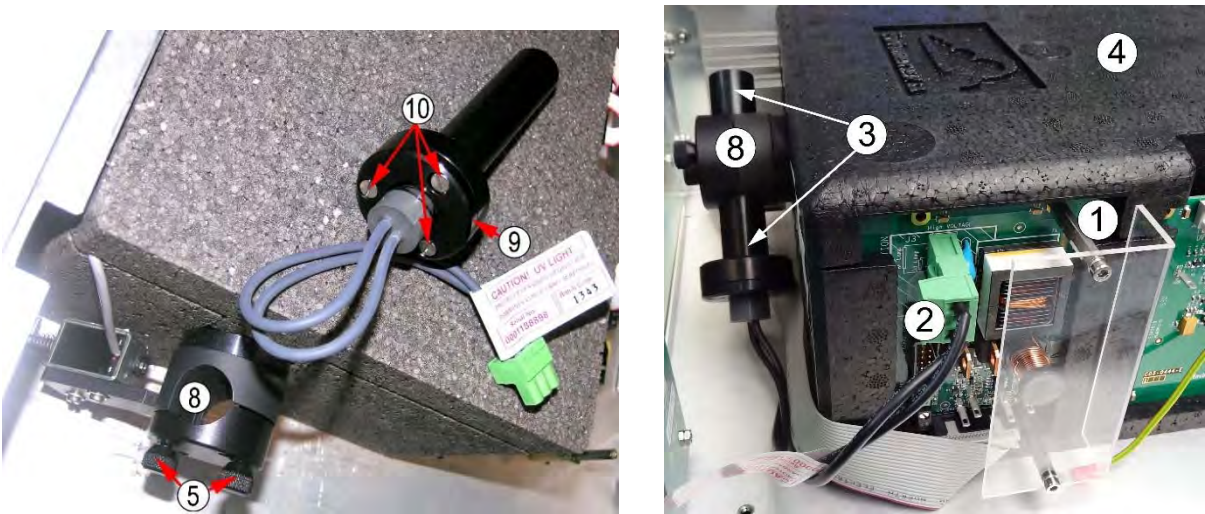
MONITOR Serial No.:	OPERATION SHEET: 4.3.5	
Scope: UV lamp replacement	PAGE: 3/3	Frequency: 2 years

WARNING: DO NOT put your fingers on the lamp glass.

UV source reassembly:

Reassembly is carried out in the reverse order from dismantling

- Replace the UV source assembly (3) in its holder (8).
- Reconnect the UV source connector (2) to the UV supply board (1).
- Plug in the power cable and power on the analyzer.
- Adjust UV source assembly position as follows :
 - Select the « *Diagnostic functions (alarm, input/output, mux...)* » screen to read the « UV Signal » parameter value.
 - Wait approximately 10-15 min for « UV signal » value stabilization.
 - Adjust the UV source assembly (3) position horizontally and vertically until reaching the maximal « UV signal » value.
 - Tighten the two UV source holder screws (5) to fix the assembly.
- Replace the analyzer cover and screw then tighten the screw located on the rear panel. Perform analyzer calibration (zero reference and span cycles).



(1) UV source supply board, (2) UV source connector, (3) UV source assembly, (4) foam block, (5) UV source holder screws, (8) UV source holder, (9) flange, (10) three screws.

Figure 4-9 – UV source reassembly

Tools and parts required

- Screwdriver, 5.0 x 75 mm
- UV lamp, Ref. : D01-1297

4.4 KITS AND SPARE PARTS SET FOR AF22e MAINTENANCE

Maintenance kit for 1 year - Ref.: AF22e-K

Line Nb	Designation	Reference	Qty
1	SO ₂ zero filter	F05-0128-C-SAV	4
2	Teflon filter (porosity 5µm - Ø 47 mm)	F05-11-842	25
3	Kit for KNF pump	V04-0005	1

Page intentionally left blank

CHAPTER 5

CORRECTIVE MAINTENANCE

Figure 5–1 – Interconnection board diagram	5–11
Figure 5–2 – Controller board diagram (above view)	5–12
Figure 5–3 – Power supply board diagram	5–13
Figure 5–4 – Measurement board diagram	5–15
Figure 5–5 – Photodiode board diagram (above view)	5–16
Figure 5–6 – PM preamplifier board diagram	5–17
Figure 5–7 – UV lamp power board diagram	5–18
Figure 5–8 – ARM20 board	5–19

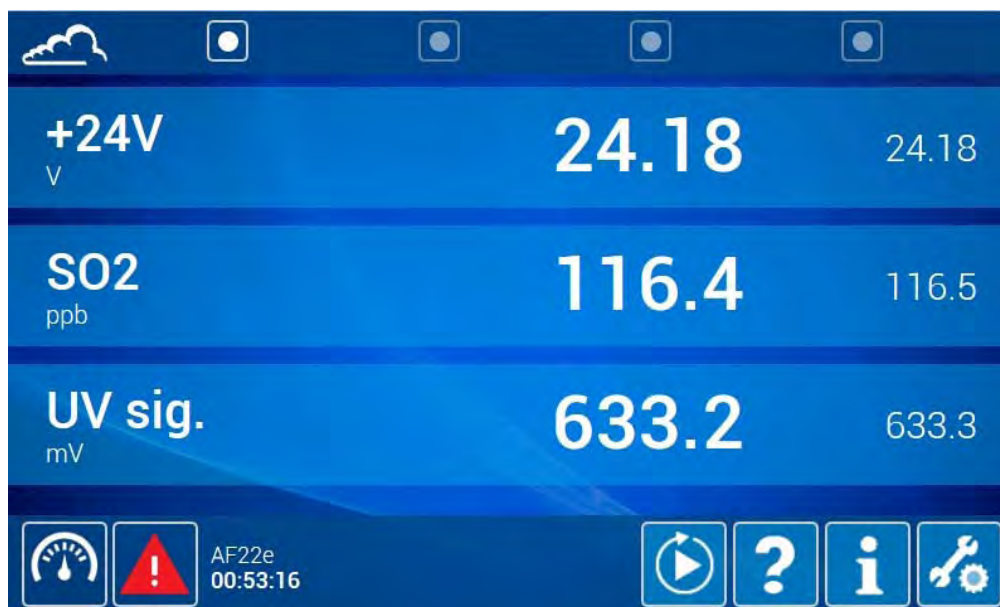
Table 5–1 – List of faults and corrective actions	5–4
Table 5–2 – Interconnection board configuration and LEDs description	5–10
Table 5–3 – Controller board configuration	5–12
Table 5–4 – Power supply board	5–13
Table 5–5 – Measurement board test points, LEDs description and connections	5–14
Table 5–6 – Photodiode board configuration	5–16
Table 5–7 – PM preamplifier board test points, LEDs description, connections and adjustments	5–17
Table 5–8 – UV lamp power board test points, LEDs description and connections	5–18
Table 5–9 – ARM20 board description	5–19

Page intentionally left blank

5. CORRECTIVE MAINTENANCE

Analyzer corrective maintenance must be carried out only by qualified people using the information provided in this document.

The monitor automatically and continuously performs self-tests of its main components. When any malfunction is detected: the alarm icon  is displayed and blinks on the home page and in the menu bar of all software screens, next to the date and analyzer name.



It is necessary to touch the alarm icon to directly access the « *DIAGNOSTIC* ⇒ *Current alarms* » screen and check the malfunction origin.

Table 5–1 summarizes the main faults indicated by the unit with their corresponding possible corrective actions.

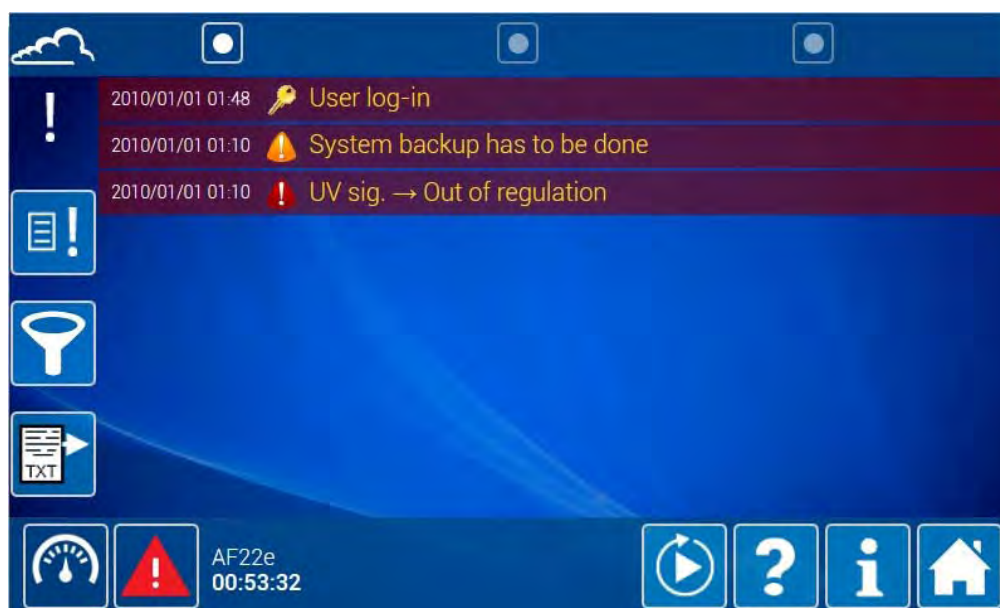


Table 5–1 – List of faults and corrective actions

Alarm message (Min./Max. values)	Possible cause(s)	Possible corrective action(s)
DAC out of range Min. value: 0 point Max. value: 4095 points	- Current measurements exceed the set DAC range. - Measurement offset is too high or too low. - Calibration factor is incorrect.	- Set a DAC range complying with the current measurements. - Check and change offset setting. - Check and correct calibration factor setting.
Out of Range Min. value: 0 ppm Max. value: 10 ppm	- SO₂ concentration exceeds the analyzer specification range. - Zero-reference cycle is not performed correctly (example: span gas is used instead of zero).	- Decrease SO₂ concentration by dilution. - Check zero filter and zero-reference cycle setting. Then, perform a new zero-reference cycle.
Zero filter outdated Min. value: Max. value: <10 days	- Internal zero filter has reached its lifetime. - Lifetime countdown of internal zero filter is incorrect.	- Replace internal zero filter and reset lifetime countdown to 180 days. - Reset lifetime countdown to 180 days.
Calibration Fault Min. value: $ \Delta K/K > 50\%$ Max. value: N.A.	- Calibration cycle was not performed with span gas. - Calibration cycle duration is too short. - Previous calibration is not performed correctly.	- Check span gas cylinder connection to analyzer span gas inlet. - In the « <i>Metrology configuration</i> » screen, check setting of « <i>Span gas value</i> » fields indicating span gas concentration connected to analyzer span gas inlet port. Then, validate former calibration coefficient value. Finally, perform a new calibration cycle if necessary. - In the « <i>Automatic cycle configuration</i> » screen, check calibration cycle duration: recommended duration is 600 sec. Change if necessary. - Calculate calibration coefficient using formula: $K_{\text{new}} = K_{\text{former}} (1 + \Delta K/K)$ and correct in « <i>Metrology configuration</i> » screen. The analyzer is now in alarm mode. Launch a new calibration cycle if necessary.

Alarm message (Min./Max. values)	Possible cause(s)	Possible corrective action(s)
No gas pressure Min. value: not present Max. value: present	– Gas pressure sensor cable is disconnected.	– Connect gas pressure sensor cable.
Gas pressure Min. value: 800 hPa Max. value: 1200 hPa	– If gas pressure is abnormally low: • Either sample, zero or span inlet ports are closed, • Or dust filter is clogged or dirty, • Or solenoid valve block is clogged or dirty, • Or gas pressure sensor is faulty. – If chamber pressure is abnormally high: • Either the analyzer's inlet port(s) gas flow is too high and the excess vent is not connected. • Or pressure sensor is faulty.	• Open sample, zero or span inlet analyzer ports. • Check dust filter and replace if necessary. • Check if the sampling line is dirty or kinked. • Check solenoid valve block status: clean with dry, oil-free, compressed air or replace if cleaning is not possible. • Replace gas pressure sensor. • Check if the excess vent is connected. • Replace pressure sensor.
Pump pressure Min. value: 100 hPa Max. value: 600 hPa	– If pump pressure is abnormally low: • Either sample, zero or span inlet ports are closed, • Or dust filter is dirty or clogged, • Or solenoid valve block is clogged or dirty, • Or pump pressure sensor is faulty.	• Open sample, zero or span inlet analyzer ports. • Check dust filter and replace if necessary. • Check if the sampling line is dirty or kinked. • Check solenoid valve block status: clean with dry, oil-free, compressed air or replace it if cleaning is not possible. • Replace pump pressure sensor.

Alarm message (Min./Max. values)	Possible cause(s)	Possible corrective action(s)
	– If pump pressure is abnormally high: • Pressure sensor is faulty. • A fluid tube is unplugged.	• Replace pressure sensor. • Check the fluid tubes.
Flow Fault Min. value: 5 l/h Max. value: 90 l/h	– Pump cable is disconnected. – A fluid tube is unplugged. – Interconnection board pump command is inactive. – Pressure sensor's inlet ports of pump and gas are disconnected.	– Connect pump cable to interconnection board. – Check the fluid tubes. – Check that interconnection board pump command is active and controller board operates correctly. – Check pump and gas pressure sensor connections.
No cham. Temp Min. value: N.A. Max. value: N.A.	– Chamber temperature sensor cable is disconnected.	– Connect chamber temperature sensor cable.
Cham. T open Min. value: N.A. Max. value: > 70°C	– Chamber temperature sensor is connected but connection is faulty (as an open circuit).	– Replace temperature sensor: analyzer exits alarm mode after 60 sec.
Cham. T shortcut Min. value: < 5°C Max. value: N.A.	– Chamber temperature sensor is connected, but connection is faulty (as a closed circuit).	– Replace chamber temperature sensor: analyzer exits alarm mode after 60 sec.
Chamber temperature Min. value: 48°C Max. value: 52°C	– Controller board configuration is faulty. – Heating laminar is not connected. – Heating laminar is faulty.	– Replace controller board with a calibrated controller board. – Connect heating laminar. – Replace heating laminar.
No internal Temp Min. value: absent Max. value: absent	– Internal temperature sensor cable is not connected.	– Connect internal temperature sensor cable.

Alarm message (Min./Max. values)	Possible cause(s)	Possible corrective action(s)
Internal temperature Min. value: 5°C Max. value: 50°C	– Ambient temperature is not within the recommended range.	– Install the analyzer in a 5-45°C thermo-controlled room.
I UV lamp Min. value: 45 mA Max. value: 49 mA	– UV lamp cable is disconnected. – Measurement board / UV lamp power board cable is disconnected. – UV lamp power board is faulty.	– Reconnect UV lamp cable. – Reconnect measurement board / UV lamp power board cable. – Replace UV lamp power board.
HV PM Min. value: 2400 mA Max. value: 3200 mA	– PM preamplifier / AF measurement board cable is disconnected. – PM preamplifier board is faulty.	– Connect PM preamplifier board / measurement board cable. – Replace PM preamplifier board.
UV ref. Min. value: 130 mV Max. value:	– UV lamp is not aligned. – UV lamp is faulty.	– Aligned UV lamp as described in chapter 4. – Replace UV lamp as described in chapter 4.
NO Controller board Min. value: absent Max. value: N.A.	– Controller board is unplugged. – Controller board is not compliant with the analyzer, or firmware is missing. – Controller board is faulty	– Turn off the analyzer and disconnect main cable. Reconnect controller board ribbon. – Update controller board. – Replace controller board.
NO Measure board Min. value: absent Max. value: N.A.	– Measurement board is disconnected. – Measurement board is not compliant with the analyzer, or firmware is missing. – Measurement board is faulty.	– Turn off analyzer and disconnect main cable. Reconnect measurement board ribbon. – Update measurement board. – Replace measurement board.

Alarm message (Min./Max. values)	Possible cause(s)	Possible corrective action(s)
Supply fault Min. value: N.A. Max. value: N.A.	– Supply board is faulty. – Measurement board microcontroller is faulty. – Abnormal power consumption is detected. – 24 V power supply is not stable.	– Replace supply board. – Replace measurement board. – Check that all devices as solenoid valves, laminar heating, shutter, or other external devices are connected to 24 V external relay output. – Replace power supply.
5 V sensor fault Min. value: 4.75 V Max. value: 5.25 V	– Power supply board is faulty. – 5 V sensors are faulty.	– Replace power supply board. – Check all pressure and temperature sensors.

Page intentionally left blank

Table 5-2 – Interconnection board configuration and LEDs description

Jumpers	Connections
J1	Controller board
J2	Power supply board
J3	Arm20 board (with ribbon cable)
J4	24 V general power supply from 230 V/24 V supply box
J5	Gas pressure sensor board
J6	Zero SV
J8	Span SV
J10	Rack fan
J15	24 V Estel board power supply, 2e series rear panel
J19	SV zero/span and SV measurement power supply, rear panel
J23	External TCP/IP LAN
J24	Internal TCP/IP
J26	UV lamp power board link
J27	AF measure board link
J28	External USB1
J29	Internal temperature
J30	External USB2
J31	Internal USB3

LED N°	Identification	LED turned on	LED turned off
LED1	Span SV	Active	Disabled
LED3	Rack fan	Rotating	Stopped
LED4	Zero SV	Active	Disabled
LED13	+ 24 V	Present	Not present

SV: solenoid valve

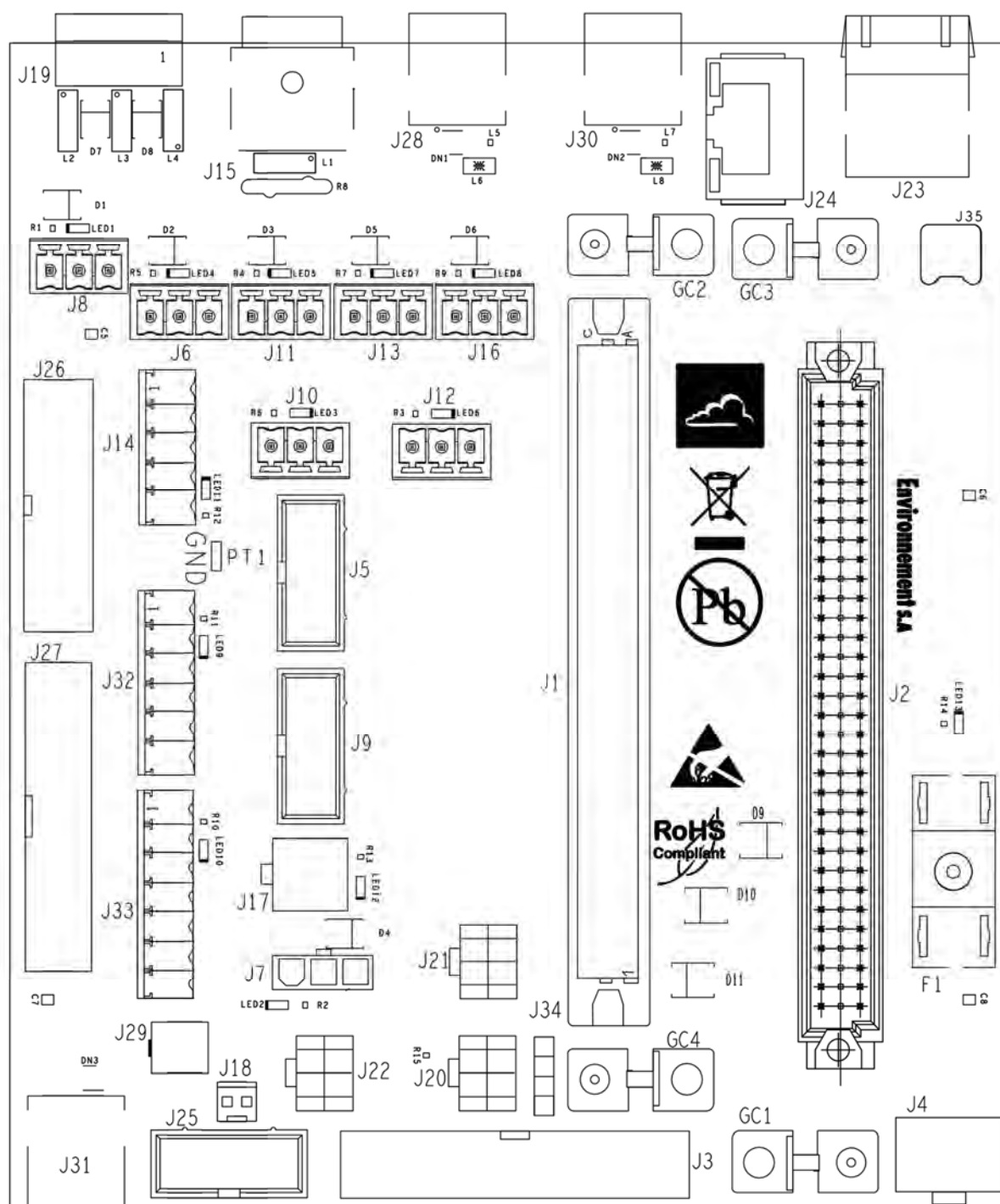


Figure 5-1 – Interconnection board diagram

Table 5-3 – Controller board configuration

Jumpers	Connections
J2	Main connector

Tests points	Signals
PT1	0 V (GND)
PT2	0 V (GND)
PT3	Power supply +24 V
PT4	Power supply +12 V
PT5	Power supply +5 V
PT6	Power supply +3,3 V

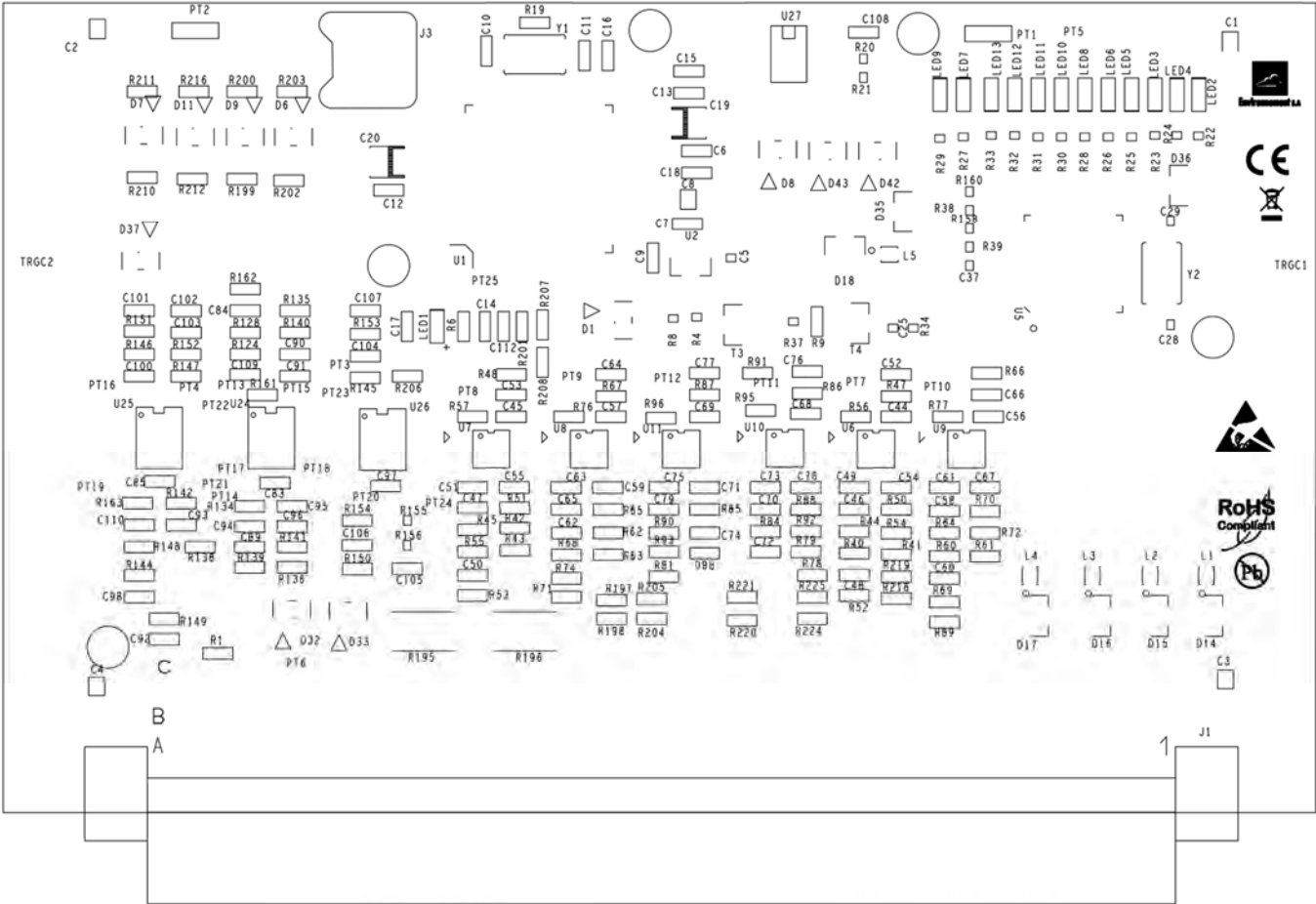


Figure 5-2 – Controller board diagram (above view)

Table 5-4 – Power supply board

Jumpers	CONNECTION
J1	Main connector

Test point	Signal
PT1	0 V (GND)

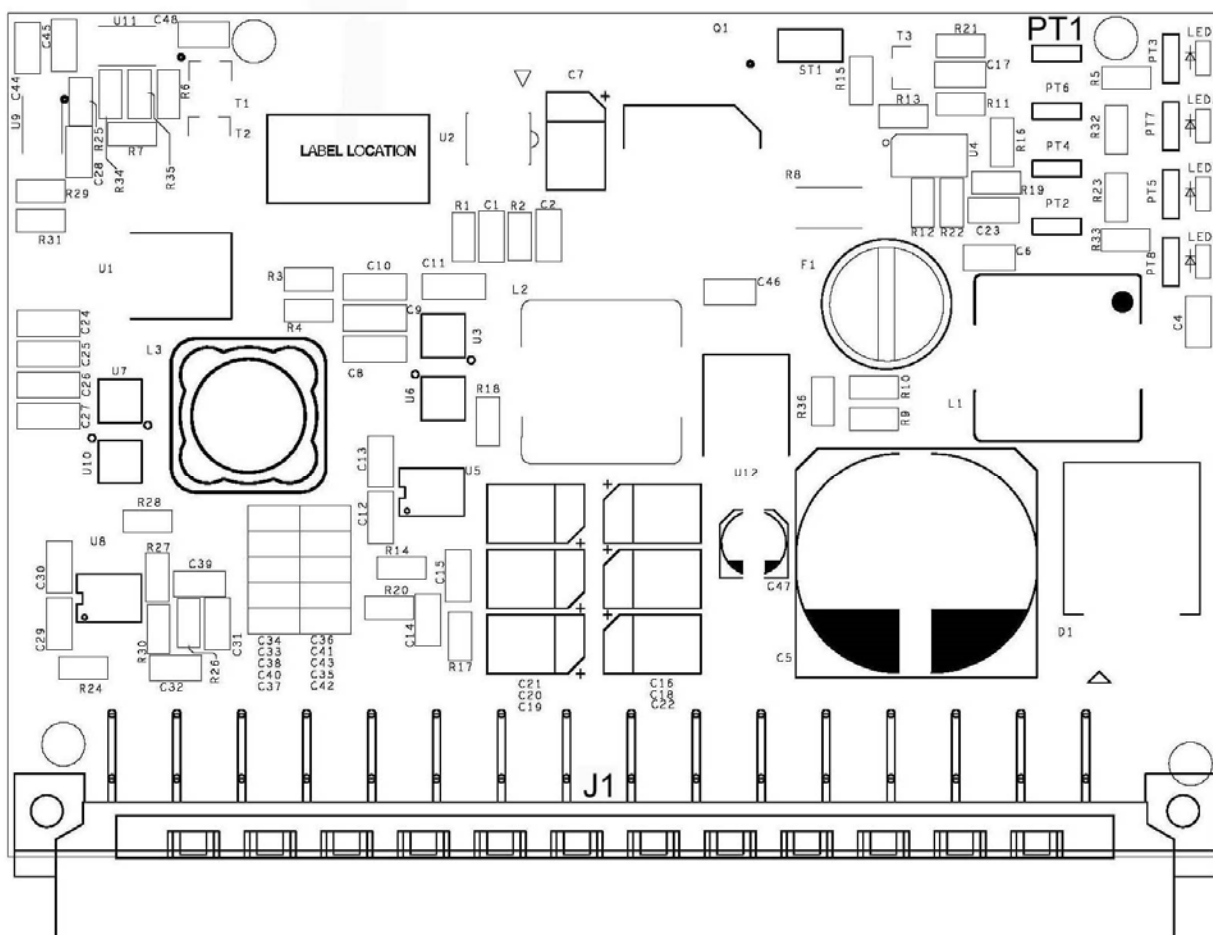


Figure 5-3 – Power supply board diagram

Table 5-5 – Measurement board test points, LEDs description and connections

Jumpers	Connections
J1	Ribbon cable towards Interconnection board
J5	Gas pressure sensor
J7	Measurement chamber temperature sensor
J8	Measurement chamber heating
J41	PM preamplifier board connector
J42	Photodiode board connector

Test points	Signal
PT1	GND
PT2	Analog GND
PT41	PM signal (high gain amplification)
PT42	UV photodiode signal
PT43	PM HV setting
PT114	PM signal (low gain amplification)

LED N°	Identification	Color	Turned on	Turned off	Blinking
LED1	USB indicator	Green	Detected	Not detected	N.A.
LED2	Board activity indicator	Orange	Active	No activity	Fast, loading mode Slow (1 Hz), normal mode
LED3	Chamber heating supply	Red	Heater ON	Heater OFF	Regulation mode ON
LED4	5 V supply	Red	Active	N.A.	N.A.

Table 5-6 – Photodiode board configuration

Jumpers	Connections
J1	Main connector

Tests points	Signals
PT1	0 V (GND)
PT2	Photodiode signal

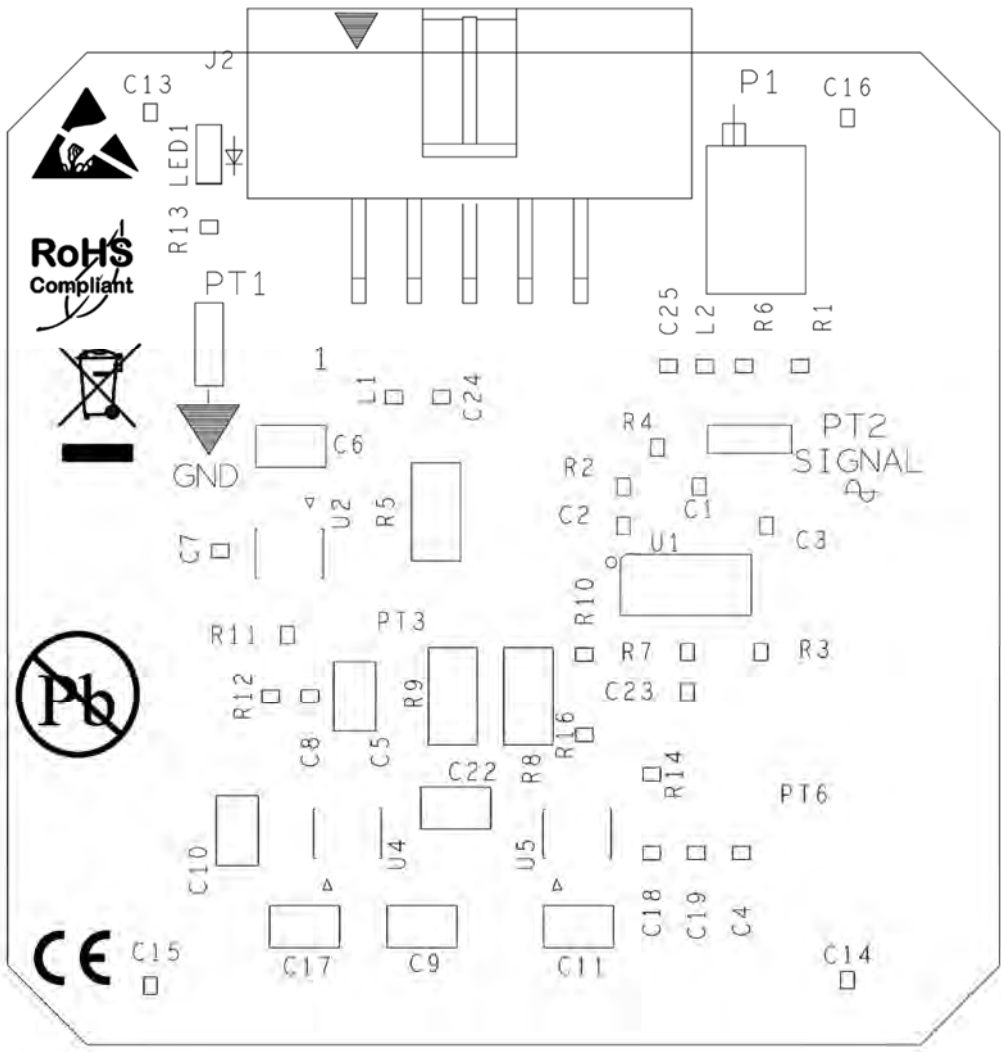


Figure 5-5 – Photodiode board diagram (above view)

**Table 5-7 – PM preamplifier board
test points, LEDs description, connections and adjustments**

Jumpers		Connections			
J1		Measurement board link			

Potentiometer*	Nature of adjustment				
P1	PM signal amplification (high gain)				
P2	PM signal offset (high gain)				
P4	PM signal amplification (low gain)				
P5	PM signal offset (low gain)				

Test points	Signal				
PT1	GND				
PT2	PM signal (high gain)				
PT6	PM signal (low gain)				

LED N°	Identification	Color	Turned on	Turned off	Blinking
LED1	5 V supply	Red	Active	N.A.	N.A.

* Turn clock wise to increase the gain. Turn counter-clock wise to decrease the gain.

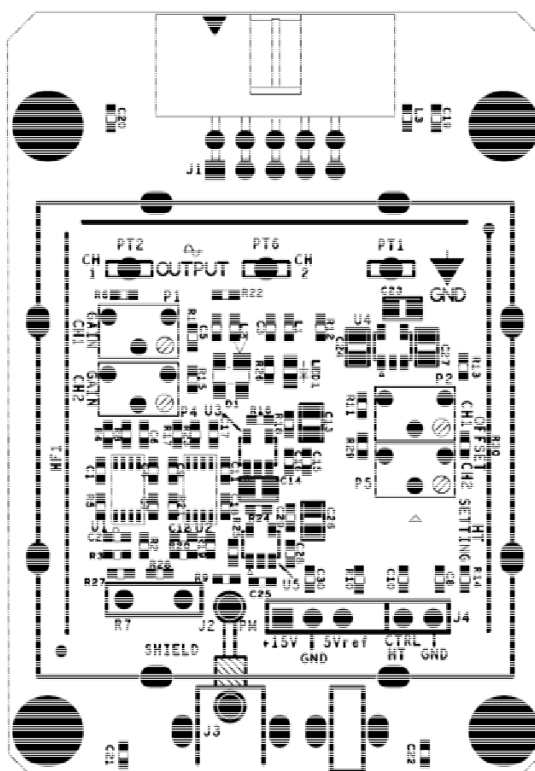


Figure 5-6 – PM preamplifier board diagram

**Table 5–8 – UV lamp power board
test points, LEDs description and connections**

Jumpers	Connections
J1	Ribbon cable to Interconnection board
J3	UV lamp connector

Test points	Signal
PT1	GND
PT2	12 V supply
PT3	5 V supply
PT5	PM setting
PT6	I lamp

LED N°	Identification	Color	Turned on	Turned off	Blinking
LED1	Lamp diagnostic	Green	Lamp OK	Lamp fault	N.A.
LED3	5 V supply	Green	Active	N.A.	N.A.
LED4	12 V supply	Green	Active	N.A.	N.A.

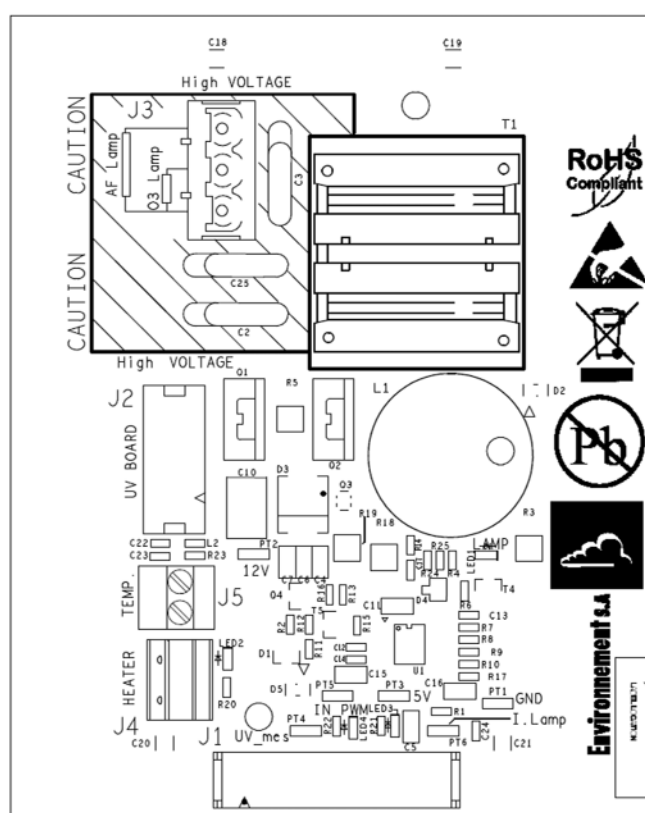


Figure 5–7 – UV lamp power board diagram

Page intentionally left blank